

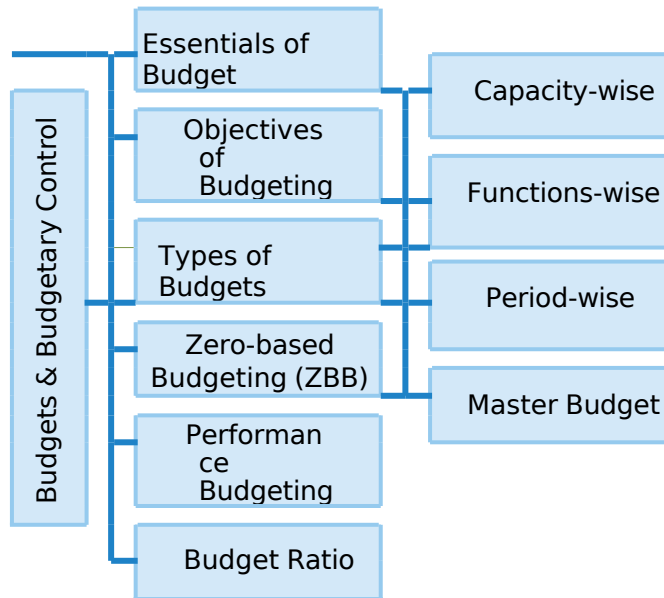
BUDGETS & BUDGETARY CONTROL



LEARNING OUTCOMES

- ♦ State the meaning of budget and budgetary control
- ♦ Essentials of budget.
- ♦ Discuss the objectives and importance of budget and budgetary control.
- ♦ Describe the process of preparing budgets.
- ♦ State the motivation in budget process.
- ♦ List the different types of budgets.
- ♦ Differentiate between fixed and flexible budget.
- ♦ Prepare fixed and flexible budget.

CHAPTER OVERVIEW



1. INTRODUCTION

An organization has its long-term objectives to achieve. The objectives are broken down into achievable goals and targets. When these goals and targets are translated into business plans, it is necessary to express the plans into quantifiable terms to make it achievable. Budget is a commonly used business language that expresses the business plans in quantifiable terms. When the targets are monitored and compared with the actual results with the objective to narrow down the deviations, make participants responsible and implement the preventive and corrective actions, is known as budgetary control.

Meaning of Budget and Budgeting

Budget: A budget is an instrument of management used as an aid in the planning, programming and control of business activity. The Chartered Institute of Management Accountants (CIMA), UK defines budget as “A financial and/or quantitative statement, prepared and approved prior to a defined period of time of the policy to be pursued during that period for the purpose of attaining a given objective. It may include income, expenditure and employment of capital” The budget is a blue-print of the projected plan of action expressed in quantitative terms for a specified period of time.

Budget and Forecast

There is some similarity between the budget and forecast as both relate to a defined period of time. A forecast is an assessment of probable future events. Budget a financial/quantitative plan of a business enterprise to be pursued over a period of time. Therefore, at the planning stage it is necessary to forecast a probable course of action for the business. Budget is a commitment or a target which the management seeks to attain on the basis of the forecasts made. Forecasts are made regarding sales, production cost and financial requirements of the business. A forecast denotes some degree of flexibility while a budget denotes a definite target.

Budgeting: Budgeting is the process of designing, implementing and operating of budget. The main emphasis in budgeting process is the provision of resources to support plans which are being implemented. It is a means of coordinating the combined intelligence of an entire organisation into a plan of action based on past performance and governed by rational judgment of factors that will influence the course of business in the future.

2. ESSENTIAL CHARACTERISTICS OF BUDGET

The main characteristics of budget are as follows:

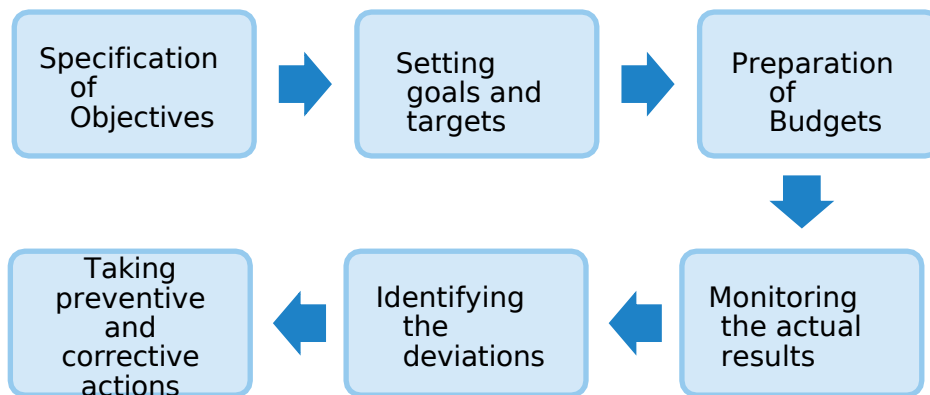
1. A budget is concerned for a definite future period.
2. A budget is a written document.
3. A budget is a detailed plan of all the economic activities of a business.
4. All the departments of a business unit should co-operate for the preparation of a business budget.
5. Budget is a means to achieve business objectives and it is not an end in itself.
6. Budget needs to be updated, corrected and controlled every time circumstances change. Therefore, it is a continuous process.
7. Budget helps in planning, coordination and control.
8. Different types of budgets are prepared by industries according to business requirements.

9. A budget acts as a business barometer.
10. Budget is usually prepared in the light of past experiences.
11. Budget is a constant endeavour of the Management.

3. ESSENTIAL STEPS FOR PREPARING BUDGET

Essential steps for preparing a budget are as follows:

1. Organisational structure must be clearly defined and responsibility should be assigned to identifiable units within the organisation.
2. Setting of clear objectives and reasonable targets. Objectives should be in consonance with the long-term plan of the organisation.
3. Objectives and responsibility should be clearly stated and communicated to the management or person responsible.
4. Budgets are prepared for the future periods based on expected course of actions.
5. Budgets are updated for the events that were not kept into the mind while establishing budgets. Hence, budgets should be flexible enough for mid-term revision.
6. The entire organisation must be committed to the preparation and implementing budgeting.
7. Budgets should be quantifiable and master budget should be broken down into various functional budgets.
8. Budgets should be monitored periodically. Variances of the actual outcomes should be compared with the actuals and variances analysed and responsibility should be fixed.
9. Budgetary performance needs to be linked effectively to the reward system.



4. OBJECTIVES OF BUDGETING



Planning

Planning is the beginning of any activity. Planning establishes the objectives of the firm and decides the course of action to achieve it. It is concerned with formulating short-term and long-term plans to achieve a particular end. Planning is a statement of what should be done, how it should be done and when it should be done. The process of preparing budget begins with the establishment of specific targets of performance and is followed by devising plans to achieve such desired goals. These targets include both the overall business targets as well as the specific targets for the individual units within the business. Establishing specific targets for future operations is part of the planning function of management, while executing actions to meet the goals is the directing function of management. It may be explained as

- Budget is prepared in synchronisation with the overall objectives of the organisation, keeping mission and corporate strategy into account. Individual plans at unit level should be in consonance with organisational plan.
- Budget reflects plans. Therefore, planning should precede the preparation of budget.
- Budgeted plans are quantified and responsibility is assigned to the persons who are responsible for execution of plan.

- Communication of business objectives through budget has helped many a company to reduce expenses during business recession.
- Planning not only motivates employees to attain goals but also improves overall decision making. During the planning phase of the budget process, all viewpoints are considered, options identified, and cost reduction opportunities assessed. This process may reveal opportunities or threats that were not known prior to the budget planning process.

Directing and Coordinating

- Once the budget plans are in place, these can be used to direct and coordinate operations in order to achieve the stated targets.
- A business, however, is much more complex and requires more formal direction and coordination.
- **The budget offers an important tool to direct and coordinate business activities and units to achieve stated targets of performance.**
- The budgetary units in an organisation are called responsibility centers. Each responsibility center is led by a manager who has the authority over and responsibility for the unit's performance.
- Objectives of each responsibility centre and degree of performance expected from them are separately communicated.

Controlling

- Control is the process of monitoring, measuring, evaluating and correcting actual results to ensure that a firm's goals and plans are achieved. Control is achieved through the process of feedback.
- As time passes, **the actual performance of an operation can be compared against the planned targets. This provides prompt feedback to employees about their performance.** If necessary, employees can use such feedback to fine-tune their activities in the future.
- Feedback received in the form of budget report from the responsibility centre is helpful to know the performance of the concerned unit.

- Any unforeseen changes into the conditions which were prevailing at the time of preparing budget are taken into account and budgets are revised to show true performance.
- Comparing actual results to the plan helps prevent unplanned expenditures. The budget helps employees to regulate their spending priorities.

The main objective of Budgeting is to help in achieving the overall objective of the organization.

5. BUDGETARY CONTROL

CIMA has defined the terms 'budgetary control' as the establishment of budgets relating to the responsibilities of executives to the requirements of a policy and the continuous comparison of actual with budgeted results, either to secure by individual action, the objective of that policy or to provide a basis for its revision. **It is the system of management control and accounting in which all the operations are forecasted and planned in advance to the extent possible and the actual results compared with the forecasted and planned results.**

5.1 Budgetary Control Involves

1. Establishment of budgets
2. Continuous comparison of actuals with budgets for achievement of targets.
3. Revision of budgets after considering the changes in the circumstances.
4. Fixation of the responsibility for failure to achieve the budget targets.

5.2 Objectives of Budgetary Control System

1. **Portraying with precision the overall aims of the business** and determining targets of performance for each section or department of the business.
2. **Laying down the responsibilities** of each of the executives and other personnel so that everyone knows what is expected of him and how he will be judged. Budgetary control is one of the few ways in which an objective assessment of executives or department is possible.

3. **Providing a basis for the comparison** of actual performance with the predetermined targets and investigation of deviation, if any, of actual performance and expenses from the budgeted figures. This naturally helps in adopting corrective measures.
4. **Ensuring optimum use of available resources** to maximise profit or production, subject to the limiting factors. Since budgets cannot be properly drawn up without considering all aspects, usually there is good co-ordination when a system of budgetary control operates.
5. **Co-ordinating various activities** of the business, and centralising control and yet enabling management to decentralise responsibility and delegate authority in the overall interest of the business.
6. **Engendering a spirit of careful forethought**, assessment of what is possible and an attempt at it. It leads to dynamism without being reckless. Of course, much depends on the objectives of the firm and the dynamism of its management.
7. **Providing a basis for revision** of current and future policies.
8. **Drawing up long range plans** with a fair measure of accuracy.
9. **Providing a yardstick** against which actual results can be compared.

5.3 Steps for establishing Budgetary Control

The following steps are necessary for establishing a good budgetary control system:

1. **Determining the objectives to be achieved**, over the budget period, and the policy or policies that might be adopted for the achievement of these objectives.
2. **Determining the activities that should be undertaken** for the achievement of the objectives.
3. **Drawing up a plan or a scheme** of operation in respect of each class of activity, in quantitative as well as monetary terms for the budget period.
4. **Laying out a system of comparison** of actual performance by each person, or department with the relevant budget and determination of causes for the variation, if any.

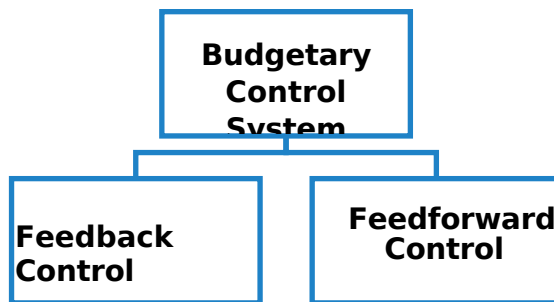
5. **Ensuring that corrective action will be taken** where the plan has not been achieved and, if that is not possible, for the revision of the plan.

In brief, it is a system to assist management in the allocation of responsibility and authority, to provide it with aid for making, estimating and planning for the future and to facilitate the analysis of the variation between estimated and actual performance.

In order to ensure effective functioning of budgetary control, it is necessary that the firm should develop a proper basis of measurement or standards with which to evaluate the efficiency of operations, *i.e.*, the firm should have in operation, a system of standard costing.

The organisation should be so integrated that all lines of authority and responsibility are properly defined. This is essential since the system of budgetary control postulates separation of functions and division of responsibilities and thus requires that the organisation shall be planned in such a manner that everyone, from the Managing Director down to the Shop Foreman, will have his duties properly defined.

5.4 Feedback and Feedforward Control



There are two types of budgetary control system based on timing of action:

Feedback Control: The feedback system of budgetary control, the actual results for the budgeted period are collected and compared with the budgeted figures. The exercise of variance identification is done after the completion of the budget period. The variances are reported and based on the report corrective actions are taken, responsibility is fixed and based on experience, modification in future targets is implemented. As the name suggests, it is an Ex-post Corrective control system of budget.

This system of budgetary control is common in organisations where Management Information System (MIS) is not so robust and where data is obtained only after the finalisation of books of account. Though this type of control system is less expensive to maintain but has limitations. Organisation has to remain on looser side in today's age of data warfare.

Feedforward Control: This is the opposite of feedback control system of budgetary control. It is Ex-Ante Preventive control mechanism of budgetary control. The budgets are set at the inception of the budgeted period and the actual results are continuously monitored and compared. The targets are kept realistic as far as possible and the targets are reviewed and reset if necessary.

This budgetary control system requires a robust MIS supported by integrated ERP system enabling an entity to get data as and when desired basis. This system is very expensive and beneficial for the organisations where the business environment is dynamic and information has important role in getting edge in competition and today's data warfare.

5.5 Budget Committee and Budget Officer

The budget committee is a group of representatives of various functions in an organisation. As all functions are inter-related and as any change in one's target will have its impact on that of the other, it is necessary to discuss the targets so that a mutually agreed programme is finally decided. This is called coordination in budget-making. It is a powerful force in knitting together various activities of the business and enforcing real control over operations.

The Chief Executive is ultimately responsible for the budget programme but it will be better if the large part of the supervisory responsibility is delegated to an official designated as Budget Officer. The budget Officer should have knowledge of the technical side of the business and should report to the president or CEO of the business entity.

The responsibility for successfully introducing and implementing Budgetary Control System rests with the Budget Committee acting through the Budget Officer. The Budget Committee would be composed of all functional heads and a member from the Board to preside over and guide the deliberations.

The main responsibilities of the Budget Committee/Budget Officer are to:

1. **Assist in the preparation of the separate budget for various departments** by coordinating the work of the accounts department, which is normally responsible to compile the budgets—with the relevant functional departments like Sales, Production, Plant maintenance etc.;
2. **Forward the budget to the individual departments heads who are re- sponsible** to implement the budget. The Budget Officer should guide them in overcoming any practical difficulties, in its working;
3. **Prepare the periodical budget reports** for circulation to the individuals concerned;
4. **Follow-up action to be taken on the budget reports;**
5. **Prepare an overall budget working report** for discussion at the Budget Committee meetings and to ensure follow-up on the lines of action suggested by the Committee;
6. **Prepare periodical reports** for the Board meeting. Comparing budgeted Profit and Loss Account and the Balance Sheet with the actual results attained.

It is necessary that every budget should be thoroughly discussed with the functional heads before it is finalised.

It is the duty of the Budget Officer to see that the periodical budget reports are supplied to the recipients at regular intervals so as to enable them to take remedial action.

The efficiency of the Budget Officer, and through him of the Budget Committee, will be judged more by the smooth working of the system and the agreement between the actual figures and the budgeted figures.

Budgets provides basis for giving an incentive for better performance,; It is up to the Budget Officer to see that attention of the different functional heads is drawn to the deviations so as to face the challenge in a successful manner.

5.6 Advantages of Budgetary Control System

Points	Description
1. Efficiency	The use of budgetary control system enables the management of a business entity to conduct its business activities in an efficient manner.
2. Control on expenditure	It is a powerful instrument used by business entity for the control of their expenditure. It provides a yardstick for measuring and evaluating the performance of individuals and their departments.
3. Finding deviations	Budget reveals the deviations of the actual from the budgeted figures after making a comparison and communicating the deviation to management.
4. Effective utilisation of resources	Effective utilisation of various resources like— men, material, machinery and money—is made possible, as the production is planned after taking these into account.
5. Revision of plans	Budget helps in the review of current trends and framing of future policies.
6. Implementation of Standard Costing system	Budget creates suitable conditions for the implementation of standard costing system in a business organisation.
7. Cost Consciousness	Budgetary control system encourages cost consciousness and maximum utilisation of available resources.
8. Credit Rating	Management which has developed a well-ordered budget plans and which operate accordingly, receive greater favour from credit agencies.

5.7 Limitations of Budgetary Control System

Points	Description
1. Based on Estimates	Budgets are based on a series of estimates, which are based on the conditions prevalent or expected at the time budget is established. It requires revision in plan if conditions change.
2. Time factor	Budgets cannot be executed automatically. Some preliminary steps are required to be accomplished before budgets are implemented. It requires proper attention and time of management. Management must not expect too much during the initial development period.
3. Co-operation Required	Staff co-operation is usually not available during the initial budgetary control exercise. In a decentralised organisation, each unit has its own objective and these units enjoy some degree of discretion. In this type of organisation structure, coordination among different units is required. The success of the budgetary control depends upon willing co-operation and teamwork,
4. Expensive	The implementation of budget is somewhat expensive. For successful implementation of the budgetary control, proper organisation structure with responsibility is prerequisite. Budgeting process start from the collection of information to for preparing the budget and performance analysis. It consumes valuable resources (in terms of qualified manpower, equipment, etc.) for this purpose; hence, it is an expensive process.
5. Not a substitute for management	Budget is only a managerial tool and must be intelligently applied for management to get benefited. Budgets are not a substitute for good management.

6. Rigid document	Budgets are sometime considered as rigid documents. But in reality, an organisation is exposed to various uncertain internal and external factors. Budget should be flexible enough to incorporate ongoing developments in the internal and external factors affecting the very purpose of the budget.
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5.8 Components of Budgetary Control System

The policy of a business for a defined period is represented by the master budget, the detailed components of which are given in a number of individual budgets called *functional budgets*. These functional budgets are broadly grouped under the following heads:

1. **Physical budgets:** Those budgets which contain information in quantitative terms such as the physical units of sales, production etc. This may include quantity of sales, quantity of production, inventories, and manpower budgets are physical budgets.
2. **Cost budgets:** Budgets which provides cost information in respect of manufacturing, administration, selling and distribution, etc. for example, manufacturing costs, selling costs, administration cost, and research and development cost budgets are cost budgets.
3. **Profit budgets:** A budget which enables the ascertainment of profit. For example, sales budget, profit and loss budget, etc.
4. **Financial budgets:** A budget which facilitates in ascertaining the financial position of a concern, for example, cash budgets, capital expenditure budget, budgeted balance sheet etc.

6 BUDGETS AND MOTIVATION

When pursuing some target, the end result of achieving the goal should be motivating one. Motivation is a factor which works like fuel to get hope lighted and ignites the aspirations. Therefore, motivation is the driving force which converts the efforts into results and thus the long-term objectives of the any person whether it would be an individual or a corporate.

The same principle of motivation also applies to a business entity to achieve its objectives in the course of pursuing its mission. Budget is a planning exercise which quantifies the desired results into targets. The budget targets are communicated to the executives at different levels and they are asked to strive to get the targets achieved. But the whole exercise is not so simple as it seems in script, implementation in practicality has bumpy rides. The behavioural aspect of human being comes into character, and it is not so difficult to guess why an executive put his/ her best efforts to achieve the communicated targets. There must be something motivating in achieving the targets, therefore, a budgeting process should have the following consideration to make it motivating one:

- (a) **Performance measurement:** The budget, at first be communicated to all executives so that everybody must be informed the desired performance expected from each of them. Secondly, the achievement of targets should have consideration in measurement and evaluation of performance an executive at individual level and at departmental level. Rewards such as promotion, increment, Performance related pay (Pay), bonus may be appropriate motivation factors.
- (b) **Achievable Targets:** While setting targets, the practical aspects such as availability of resources and realism of figures must be considered. The targets should be balance one, it neither be very easy nor too tough, means it should be realistic one. An unrealistic target has reverse impact and may be demotivate the executives.
- (c) **Optimum utilisation of resources:** A budget targets which is easily achievable may underutilise the resources such as potential skills of executives. Pressure sometime forcing to explore innovative ways to get things done. Thus, to keep motivation alive, a balanced approach should be applied for optimum utilisation of resources upto its effort zone, though beyond the comfort zone.
- (d) **Involvement in budgeting process:** The budgets which involves the executives from all department can capture the requirement of all the users. The participative budgeting motivates the executives and give them a sense of ownership. Involvement at planning stage of budget can take care of the requirements of the executives and force them accept the targets. However, involvement at every stage of budgeting process may distort the objective of budget and lands nowhere., thus, a balance approach may be followed.



7. PREPARATION OF BUDGETS

1. **Defining business or organisational objectives:** A budget is a plan for the achievement of certain organisational objectives. It is therefore desirable that these objectives are defined precisely. The organisational objectives should be written down; the areas of control demarcated; and items of revenue and expenditure to be covered by the budget clearly stated. This will give a clear understanding of the plan and its scope to all those who must cooperate to make it successful.
2. **Identification of the key budget factor:** There are usually one or two key budget factors (sometimes there may be more than two) which set a limit to the total activity. For instance, in India sometimes non-availability of power does not allow production to increase in spite of heavy demand. Similarly, lack of demand may limit production. Such a factor is known as key factor. For proper budgeting, it must be identified and its influence on production on sales estimated properly while preparing the budget.
3. **Appointment of controller/officer:** Formulation of a budget usually requires service of a whole time senior executive.; He must be assisted in this work by a Budget Committee, consisting of all the heads of departments along with the Managing Director as the Chairman. The Budget Controller/Officer is responsible for coordinating and development of budget programmes and preparing the manual of instruction, known as Budget manual.
4. **Budget Manual:** The budget manual is a booklet specifying the objectives of an organisation in relation to its strategy. The budget is made to decide how much an organisation would earn and spend and in what manner. In the budget, the organisation sets its priorities too.

CIMA, London, defines budget manual as, "A document which sets out the responsibilities of the persons engaged in, the routine of, and the forms and records required for, budgetary control."

Effective budgetary planning relies on the provision of adequate information to the individuals involved in the planning process. Many of these information needs are contained in the budget manual. A budget manual is a collection of documents that contains key information for those involved in the planning process.

Contents of a Budget Manual

Typical budget manual may include the following:

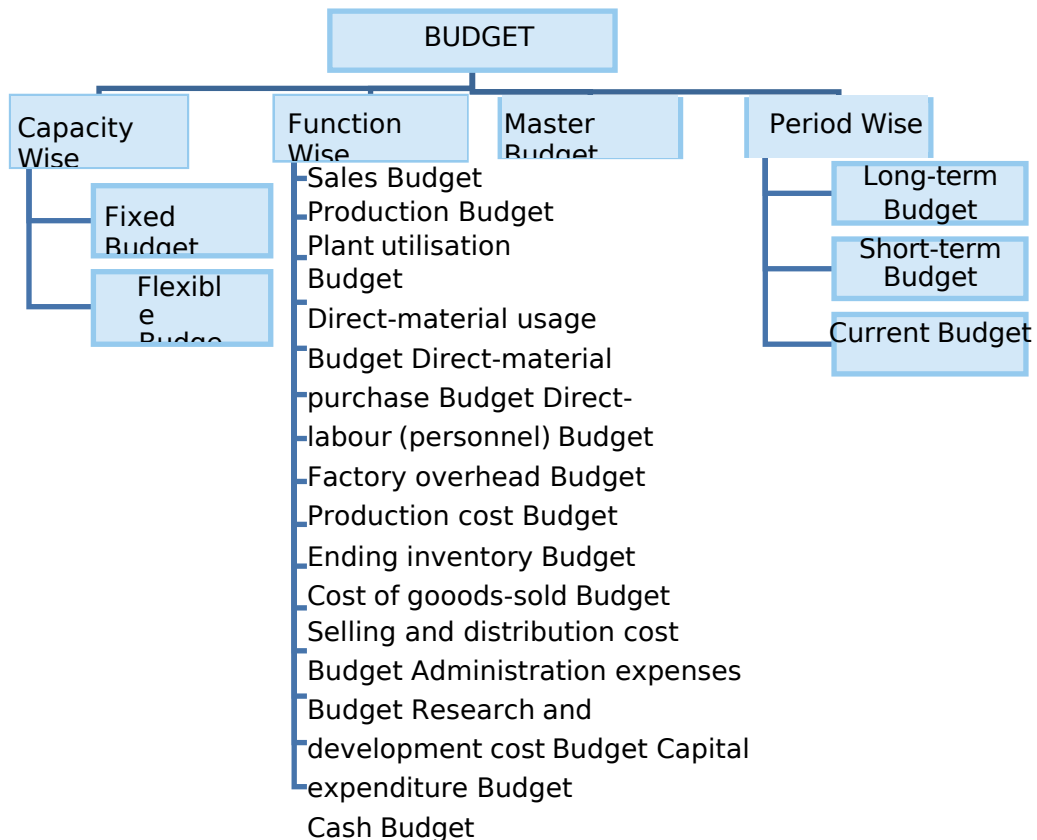
- (i) A statement regarding the objectives of the organisation and how they can be achieved through budgetary control;
- (ii) A statement about the functions and responsibilities of each executive, both regarding preparation and execution of budgets;
- (iii) Procedures to be followed for obtaining the necessary approval of budgets. The authority of granting approval should be stated in explicit terms. Whether, one two or more signatures are required on each document should be clearly stated;
- (iv) A form of organisation chart to show who are responsible for the preparation of each functional budget and the way in which the budgets are interrelated.
- (v) A timetable for the preparation of each budget.
- (vi) The manner of scrutiny and the personnel to carry it out;
- (vii) Reports, statements, forms and other record to be maintained;
- (viii) The accounts classification to be employed. It is necessary that the framework within which the costs, revenue and other financial accounts are classified must be identical both in the accounts and budget department;
- (ix) The reporting of the remedial action;
- (x) The manner in which budgets, after acceptance and issuance, are to be revised or the matter amended these are included in budgets and on which action can be taken only with the approval of top management

- (xi) This will prevent the formation of a 'bottleneck' with the late preparation of one budget holding up the preparation of all others.
 - (xii) Copies of all forms to be completed by those responsible for preparing budgets, with explanations concerning their completion.
 - (xiii) A list of the organization's account codes, with full explanations of how to use them.
 - (xiv) Information concerning key assumptions to be made by managers in their budgets, for example the rate of inflation, key exchange rates, etc.
5. **Budget period:** The period covered by a budget is known as budget period. There is no general rule governing the selection of the budget period. In practice the Budget Committee determines the length of the budget period suitable for the business. Normally, a calendar year or a period co-terminus with the financial year is adopted. The budget period for the calendar or financial year is then divided into shorter periods; it may be monthly or quarterly or for such periods as coincide with period of trading activity of the business.
6. **Standard of activity or output:** For preparing budgets for the future, past statistics, though important, cannot be completely relied upon. The past usually represents a combination of good and bad factors. Therefore, though results of the past should be studied, but these should only be applied when there is a likelihood of similar conditions repeating in the future. Also, while setting the targets for the future, it must be remembered that in a progressive business, the achievement of a year should normally exceed those of earlier years. Therefore, what was good in the past is only fair for the current year and should work for much better in the future.

In budgeting, fixing the budget of sales, expenses, and of capital expenditure is important since these budgets determine the extent of development activity. For budgeting sales, one must consider the trend of economic activity of the country, recommendations of salesmen, customers and employees, effect of price changes on sales, the provision for advertisement campaign plan capacity etc.



8. DIFFERENT TYPES OF BUDGETS



8.1 Classification on the basis of Capacity or Flexibility

These types of budgets are prepared on the basis of activity level or utilization of capacity. These are also known as “Budgets on the basis of flexibility”.

(i) Fixed Budget: A budget prepared on the basis of standard or fixed level of activity is known as fixed budget. It does not change with a change in the level of activities. According to CIMA, “**a fixed budget is a budget designed to remain unchanged irrespective of the level of activity actually attained**”. A fixed budget shows the expected results of a responsibility center for only one activity level.

Once the budget is prepared, it is not changed, even if the level of activity changes. Fixed budgeting is used by many service companies and for some administrative

functions of manufacturing companies, such as purchasing, engineering, and accounting.

Fixed Budget is used as an effective tool of cost control. In case, the level of activity attained is different from the level of activity for budgeting purposes, the fixed budget becomes ineffective. Fixed budget is suitable for fixed expenses. It is also known as a static budget.

Essential conditions:

1. When the nature of business is not seasonal.
2. There is no impact of external factors on the business activities.
3. The demand of the product is certain and stable.
4. Supply orders are received and issued regularly.
5. The market of the product is normally domestic but it can also apply in respect of service export, where fairly regular export orders are received
6. There is no need of special labour or material in the production of the products.
7. Supply of production inputs is regular.
8. There is a trend of price stability.

Generally, all above conditions are not found in practice. Hence fixed budget is not suitable in business concerns.

Merits and Demerits of fixed budgets are tabulated below:

Merits	Demerits
1. Very simple to understand 2. Less time consuming	1. It does not suite a dynamic organization and may give misleading results. A poor or good performance may remain un-noticed. 2. It is not suitable for long period. 3. It is also found unsuitable particularly when the business conditions are changing constantly. 4. Accurate estimates are not possible.

(ii) Flexible Budget: A flexible budget is a budget which, by recognising the difference in behaviour between fixed and variable costs in relation to fluctuations in output, turnover, or other variable factors, is designed to change appropriately with such fluctuations. According to CIMA, **“a flexible budget is defined as a budget which, by recognizing the difference between fixed, semi-variable and variable costs is designed to change in relation to the level of activity attained.”** Unlike static (fixed) budgets, the flexible budgets show the expected results of a responsibility center for different activity levels.

One can view a flexible budget as a series of static budgets for different levels of activity. Such budgets are especially useful in estimating and controlling factory costs and operating expenses. It is more realistic and practicable because it gives due consideration to behaviour of revenue and cost at different levels of activity. While preparing a flexible budget, the expenses are classified into three categories viz.

- (i) Fixed,
- (ii) Variable, and
- (iii) Semi-variable.

Semi-variable expenses are further segregated into fixed and variable expenses. Flexible budgeting may be resorted to under the following situations:

- (i) In the case of new business venture, due to its typical nature, it may be difficult to forecast the demand of a product accurately.
- (ii) Where the business is dependent upon the fluctuations of nature e.g., a person dealing in wool trade may have enough market demand, if temperature goes below the freezing point and much less demand if the weather is relatively warm.
- (iii) In the case of labour-intensive industry where the production of the entity is dependent upon the availability of labour.

Suitability for flexible budget:

1. Seasonal fluctuations in sales and/or production, for example in soft drinks industry;

2. a company which keeps on introducing new products or makes changes in the design of its products frequently;
3. industries engaged in make-to-order business like ship building;
4. an industry which is influenced by changes in fashion; and
5. general changes in sales.

Merits and Demerits of flexible budgets are tabulated below:

Merits	Demerits
1. With the help of flexible budget, the sales, costs and profit may be calculated easily by the business at various levels of production capacity. 2. In flexible budget, adjustment is very simple according to change in business conditions. 3. It also helps in determination of production level as it shows budgeted costs with classification at various levels of activity along with sales. Hence the management can easily select the level of production which shows the profit predetermined by the owners of the business. 4. It also shows the quantity of product to be produced to earn determined profit.	1. The formulation of flexible budget is possible only when there is proper accounting system maintained, perfect knowledge about the factors of production and various business circumstances is available. 2. Flexible Budget also requires the system of standard costing in business. 3. It is very expensive and labour oriented.

Difference between Fixed and Flexible Budgets:

Sl. No.	Fixed Budget	Flexible Budget
1.	It does not change with actual volume of activity achieved. Thus, it is known as rigid or inflexible budget.	It can be re-casted on the basis of activity level to be achieved. Thus, it is not rigid.

2.	It operates on one level of activity and under one set of conditions. It assumes that there will be no change in the prevailing conditions, which is unrealistic.	It consists of various budgets for different levels of activity.
3.	Here as all costs like - fixed, variable and semi-variable are related to only one level of activity so variance analysis does not give useful information.	Here analysis of variance provides useful information as each cost is analysed according to its behaviour.
4.	If the budgeted and actual activity levels differ significantly, then the aspects like cost ascertainment and price fixation do not give a correct picture.	Flexible budgeting at different levels of activity facilitates the ascertainment of cost, fixation of selling price and tendering of quotations.
5.	Comparison of actual performance with budgeted targets will be meaningless specially when there is a difference between the two activity levels.	It provides a meaningful basis of comparison of the actual performance with the budgeted targets.

ILLUSTRATION 1

A factory which expects to operate 7,000 hours, i.e., at 70% level of activity, furnishes details of expenses as under:

<i>Variable expenses</i>	<i>`1,260</i>
<i>Semi-variable expenses</i>	<i>`1,200</i>
<i>Fixed expenses</i>	<i>`1,800</i>

The semi-variable expenses go up by 10% between 85% and 95% activity and by 20% above 95% activity. PREPARE a flexible budget for 80, 90 and 100 per cent activities.

SOLUTION

Head of Account	Control basis	70%	80%	90%	100%
Budgeted hours		7,000	8,000	9,000	10,000
		([₹])	([₹])	([₹])	([₹])
Variable expenses	Variable	1,260	1,440	1,620	1,800
Semi-variable expenses	Semi-variable	1,200	1,200	1,320	1,440
Fixed expenses	Fixed	1,800	1,800	1,800	1,800
Total expenses		4,260	4,440	4,740	5,040
Recovery rate per hour:					
Total expenses/Bud hours		0.61	0.55	0.53	0.50

Conclusion:

We notice that the recovery rate at 70% activity is ` 0.61 per hour. If in a particular month the factory works 8,000 hours, it will be incorrect to estimate the allowance as `4,880 @ `0.61. The correct allowance will be `4,440 as shown in the table. If the actual expenses are `4,500 for this level of activity, the company has not saved any money but has over-spent by `60 (`4,500 - `4,440).

ILLUSTRATION 2

A department of Company X attains sale of ` 6,00,000 at 80 per cent of its normal capacity and its expenses are given below:

<i>Administration costs:</i>	([₹])
<i>Office salaries</i>	90,000
<i>General expenses</i>	2 per cent of sales
<i>Depreciation</i>	7,500
<i>Rates and taxes</i>	8,750
<i>Selling costs:</i>	
<i>Salaries</i>	8 per cent of sales
<i>Travelling expenses</i>	2 per cent of sales
<i>Sales office expenses</i>	1 per cent of sales
<i>General expenses</i>	1 per cent of sales

Distribution costs:

Wages	15,000
Rent	1 per cent of sales
Other expenses	4 per cent of sales

PREPARE flexible administration, selling and distribution costs budget, operating at 90 per cent, 100 per cent and 110 per cent of normal capacity.

SOLUTION

Flexible Budget of Department of Company 'X'

	80% (₹)	90% (₹)	100%(₹)	110%(₹)
Sales	6,00,000	6,75,000	7,50,000	8,25,000
Administration Costs:				
Office Salaries (fixed)	90,000	90,000	90,000	90,000
General expenses (2% of Sales)	12,000	13,500	15,000	16,500
Depreciation (fixed)	7,500	7,500	7,500	7,500
Rent and rates (fixed)	8,750	8,750	8,750	8,750
(A) Total Adm. Costs	1,18,250	1,19,750	1,21,250	1,22,750
Selling Costs:				
Salaries (8% of sales)	48,000	54,000	60,000	66,000
Travelling expenses (2% of sales)	12,000	13,500	15,000	16,500
Sales office (1% of sales)	6,000	6,750	7,500	8,250
General expenses (1% of sales)	6,000	6,750	7,500	8,250
(B) Total Selling Costs	72,000	81,000	90,000	99,000
Distribution Costs:				
Wages (fixed)	15,000	15,000	15,000	15,000
Rent (1% of sales)	6,000	6,750	7,500	8,250
Other expenses (4% of sales)	24,000	27,000	30,000	33,000
(C) Total Distribution Costs	45,000	48,750	52,500	56,250
Total Costs (A + B + C)	2,35,250	2,49,500	2,63,750	2,78,000

Note: In the absence of information, it has been assumed that office salaries, depreciation, rates and taxes and wages remain the same at 110% level of activity also. However, in practice some of these costs may change if present capacity is exceeded.

ILLUSTRATION 3

Action Plan Manufacturers normally produce 8,000 units of their product in a month, in their Machine Shop. For the month of January, they had planned for a production of 10,000 units. Owing to a sudden cancellation of a contract in the middle of January, they could only produce 6,000 units in January.

Indirect manufacturing costs are carefully planned and monitored in the Machine Shop and the Foreman of the shop is paid a 10% of the savings as bonus when in any month the indirect manufacturing cost incurred is less than the budgeted provision.

The Foreman has put in a claim that he should be paid a bonus of ` 88.50 for the month of January. The Works Manager wonders how anyone can claim a bonus when the Company has lost a sizeable contract. The relevant figures are as under:

Indirect manufacturing	Expenses for a normal month (`)	Planned for January (`)	Actual in costs January (`)
Salary of foreman	1,000	1,000	1,000
Indirect labour	720	900	600
Indirect material	800	1,000	700
Repairs and maintenance	600	650	600
Power	800	875	740
Tools consumed	320	400	300
Rates and taxes	150	150	150
Depreciation	800	800	800
Insurance	100	100	100
	5,290	5,875	4,990

Do you agree with the Works Manager? Is the Foreman entitled to any bonus for the performance in January? Substantiate your answer with facts and figures. EXPLAIN.

SOLUTION

Flexible Budget of “Action Plan Manufacturers” (for the month of January)

Indirect manufacturing cost	Nature of cost	Expense s for a normal month (`) (2)	Planned expense s (`) (3)	Expense s as per flexible budget (`) (4)	Actual expens es (`) (5)	Difference (`) (6)=(5) - (4)
Salary of foreman	Fixed	1,000	1,000	1,000	1,000	Nil
Indirect labour (WN 1)	Variable	720	900	540	600	60
Indirect material (WN 2)	Variable	800	1,000	600	700	100
Repair and maintenance (WN 3)	Semi-variable	600	650	550	600	50
Power (WN 4)	Semi-variable	800	875	725	740	15
Tools consumed (WN 5)	Variable	320	400	240	300	60
Rates and taxes	Fixed	150	150	150	150	Nil
Depreciation	Fixed	800	800	800	800	Nil
Insurance	Fixed	100	100	100	100	Nil
		5,290	5,875	4,705	4,990	285

Conclusion: The above statement of flexible budget shows that the concern’s expenses in the month of January have increased by `285 as compared to flexible budget. Under such circumstances, assuming the expenses are controllable and based on the financial perspective the Foreman of the company should not be entitled for any performance bonus for the month of January.

Working notes:

1. Indirect labour cost per unit $\frac{8,000}{720} = ₹ 0.09$

Indirect labour for 6,000 units = 6,000 × ₹ 0.09 = ₹ 540.

2. Indirect material cost per unit $\frac{800}{8,000} = ₹ 0.10$

Indirect material for 6,000 units = 6,000 × ₹ 0.10 = ₹ 600

3. According to high and low point method of segregating semi-variable cost into fixed and variable components, following formulae may be used.

$$\text{Variable cost of repair and maintenance per unit} = \frac{\text{Change in expense level}}{\text{Change in output level}}$$
$$= \frac{₹ 650 - ₹ 600}{2,000} = ₹ 0.025$$

For 8,000 units

Total Variable cost of repair and maintenance =
₹ 200 Fixed repair & maintenance cost =
₹ 400

Hence at 6,000 units output level, total cost of repair and maintenance should be

$$= ₹ 400 + ₹ 0.025 \times 6,000 \text{ units} = ₹ 400 + ₹ 150 = ₹ 550$$

4. Variable cost of power per unit = $\frac{₹ 875 - ₹ 800}{2,000 \text{ units}} = ₹ 0.0375$

For 8,000 units

Total variable cost of power =
₹ 300 Fixed cost = ₹ 500

Hence, at 6,000 units output level, total cost of power should be

$$= ₹ 500 + ₹ 0.0375 \times 6,000 \text{ units} = ₹ 500 + ₹ 225 = ₹ 725$$

5. Tools consumed cost for 8,000 units = ₹320

Hence, tools consumed cost for 6,000 units = (₹320/8,000 units) × 6,000 units
= ₹240

8.2 Classification on the basis of Function

A functional budget is one which is related to function of the business as for example, production budget relating to the manufacturing function. Functional budgets are prepared for each function and they are subsidiary to the master budget of the business.

The various types of functional budgets to be prepared will vary according to the size and nature of the business.

The various commonly used functional budgets are:

- (i) Sales Budget
- (ii) Production Budget
- (iii) Plant Utilisation Budget
- (iv) Direct-Material Usage Budget
- (v) Direct-Material Purchase Budget
- (vi) Direct Labour (Personnel) Budget
- (vii) Production or Factory Overhead Budget
- (viii) Production Cost Budget
- (ix) Ending Inventory Budget
- (x) Cost of Goods Sold Budget
- (xi) Selling and Distribution Cost budget
- (xii) Administration Expenses Budget
- (xiii) Research and Development Cost Budget
- (xiv) Capital Expenditure Budget
- (xv) Cash Budget

The important functional budgets (also known as schedules to Master Budget) and the master budget are discussed and illustrated below:

(i) Sales Budget:

- **Sales forecast is the commencement of budgeting and hence sales budget assumes primary importance.** The quantity which can be sold may be the principal budget factor in many business undertakings. In any case in order to chalk out a realistic budget programme, there must be an accurate sales forecast.
- The sales budget is prepared for each product. This includes:
 1. the quantity of estimated sales and
 2. the expected unit selling price. These data are often reported by regions or by sales representatives.
- In estimating the quantity of sales for each product, past sales volumes are often used as a starting point. These amounts are adjusted (increased or decreased) for factors that are expected to affect future sales. Such as the factors listed below.
 - (i) Backlog of unfulfilled sales orders
 - (ii) Planned advertising and promotion
 - (iii) Expected industry and general economic conditions
 - (iv) Productive capacity
 - (v) Projected pricing
 - (vi) Findings of market research studies
 - (vii) Relative product profitability.
 - (viii) Competition.
- Once an estimate of the sales volume is obtained, **the expected sales revenue can be determined by multiplying the volume by the expected unit sales price.** The sales budget represents the total sales in physical quantities and values for a future budget period. Sales managers are constantly faced with problems like anticipation of customer requirements, new product needs, competitor strategies and various changes in distribution methods or promotional techniques.

Product Y										
1st Qtr.										
Total										

Example of sales budget:

XYZ COMPANY

Sales Budget for the year ending March, 20....

	Units	Selling price Per unit (₹)	Total (₹)
Product A	5,000	75	3,75,000
Product B	10,000	80	8,00,000
			11,75,000

(ii) Production Budget:

Production Budget is a forecast of the production for the budget period of an organisation. Production budget is prepared in two parts, viz. production volume budget for the physical units of the products to be manufactured and the cost of production or manufacturing budget detailing the budgeted cost under material, labour, and factory overhead in respect of the products. Production budget shows the production for the budget period based upon:

1. Sales budget,
2. Production capacity of the factory,
3. Planned increase or decrease in finished stocks, and
4. Policy governing outside purchase.

Production budget is normally stated in units of output. Production should be carefully coordinated with the sales budget to ensure that production and sales are kept in balance during the period. **The number of units to be manufactured to meet budgeted sales and inventory needs for each product is set forth in the production budget.**

The production facility available and the sales budget will be compared and coordinated to determine the production budget. If production facilities are not sufficient, consideration may be given to such factors as working overtime, introducing shift working, sub-contracting or purchasing of

additional plant and machinery. If, however, the production facilities are surplus, consideration should be given to promote advertising, reduction of prices to increase the sales, sub-contracting of surplus capacity, etc.

One of the conditions to be considered in all the compilation of production budget is the level of stock to be maintained.

- **The level of stocks will depend upon the following three factors viz.:**
 1. Seasonal industries in which stocks have to be built up during off season to cater to the peak season,
 2. A steady and uniform level of production to utilise the plant fully and to avoid retrenchment or lay-off of the workers, and
 3. To produce in such a way that minimum stocks are maintained at any time to avoid locking up of funds in inventory.
- **Production budget can, therefore, show:**
 1. Stabilised production every month, say, the maximum possible production or
 2. Stabilised minimum quantity of stocks which will reduce inventory costs.
 3. In the case of stabilised production, the production facility will be fully utilized, but the inventory carrying costs will vary according to stocks held. In the case of stabilised stocks method, however, the inventory carrying will be the lowest, but there may be under-utilisation of capacity.

Example of production budget:

XYZ COMPANY

Production budget in units for the year ending March 31, 20....

	Products	
	A	B
Budgeted sales	5,000	10,000
Add : Desired closing stock	500	1,000
Total quantity required	5,500	11,000

Less : Opening stock	1,500	2,000
Units to be produced	4,000	9,000

(iii) Plant Utilisation Budget:

Plant utilisation budget represents, in terms of working hours, weight or other convenient units of plant facilities required to carry out the programme laid down in the production budget.

The main purposes of this budget are:

1. To determine the load on each process, cost or groups of machines for the budget period.
2. To indicate the processes or cost centres which are overloaded so that corrective action may be taken such as: (i) working overtime (ii) sub- contracting (iii) expansion of production facility, etc.
3. To dovetail the sales production budgets where it is not possible to increase the capacity of any of the overloaded processes.
4. Where surplus capacity is available in any of the processes, to make effort to boost sales to utilise the surplus capacity.

(iv) Direct Material usage Budget:

The steps involved in the compilation of direct materials usage budget are as under:

1. The quality standards for each item of material have to be specified. In this connection, standardisation of size, quality, colour, etc., may be considered.
2. Standard requirement of each item of materials required should also be set. While setting the standard quality, consideration should be given to normal loss in process. The standard allowance for normal loss may be given on the basis of past performance, test runs, technical estimates etc.
3. Standard prices for each item of materials should be set after giving consideration to stock and contracts entered into.

After setting standards for quality, quantity and prices, the direct materials cost budget can be prepared by multiplying each item of material required for the production by the standard price.

Example of direct material usage budget is as under:

XYZ COMPANY Direct material usage in units and in amount for the year ending March 31, 20...					
	Direct Materials				
Type of material	Product A (4,000 units)	Product B (9,000 units)	Total direct material usage (Units)	Material cost per unit (₹)	Total cost of material used (₹)
X (12 units per finished product)	48,000	1,08,000	1,56,000	1.50	2,34,000
Y (4 units per product A & 2 units per product B)	16,000	18,000	34,000	2.50	85,000
				Total	3,19,000

(v) Direct Material Purchase Budget:

- **The production budget is the starting point for determining the estimated quantities of direct materials to be purchased.**
- Multiplying these quantities by the expected unit purchase price determines the total cost of direct materials to be purchased.

Two important considerations that govern purchase budgets are as follows:

- Economic order quantity.
 - Re-order point with safety stocks to cover fluctuations in demand.
- The direct material purchases budget helps management maintain inventory levels within reasonable limits. For this purpose, the timing of the direct materials purchases should be

coordinated between the purchasing and production departments.

An example of material purchase budget is as under:

XYZ Company			
Direct material purchase budget for the year ending March 31, 20.....			
	Material X	Material Y	Total
Desired closing stock (units)	3,000	500	
Units required for production	1,56,000	34,000	
<i>Add:</i>			
Total Requirement	1,59,000	34,500	
<i>Less: Opening stock (units)</i>	4,000	300	
Units to be purchased	1,55,000	34,200	
Unit price (`)	1.50	2.50	
Purchase cost (`)	2,32,500	85,500	3,18,000

(vi) Direct Labour (Personnel) Budget:

- Once sales budget and Production budget are compiled and plant utilisation budget is decided detailed amount of the various machine operations involved and services required can be calculated. This will facilitate preparation of an estimate of different grades of labour required.

From this, the standard hours required to be worked can be calculated the total labour component thus budgeted can be divided into direct and indirect labour. Standard rates of wages for each grade of labour can be introduced and then the direct and indirect labour cost budget can be prepared.

Merits/advantages:

1. It defines the direct and indirect labour force required.
2. It enables the personnel department to plan ahead in recruitment and training of workers so that labour turnover can be reduced to the minimum.
3. It reveals the labour cost to be incurred in the manufacture, to facilitate preparation of manufacturing cost budgets and cash budgets for financing the wage bill.

Example of direct-labour cost budget:

XYZ COMPANY Direct-labour cost budget for the year ending March 31, 20...				
	Units to be produced	Direct labour hour, per unit	Total hours	Total budget cost (₹) @ ₹ 2 per hour
Product A	4,000	7	28,000	56,000
Product B	9,000	10	90,000	1,80,000
			1,18,000	2,36,000

(vii) Production or Factory Overhead Budget:

- Production overheads consist of all items such as indirect materials, indirect labour and indirect expenses. Indirect expenses. These include expenditures on factors such as power, fuel, fringe benefits, depreciation etc. The estimated overheads which are necessary for production in the factory are called factory overhead costs and included in the factory overhead budget.
- Factory overhead budget usually includes the total estimated cost for each item of factory overhead.
- The production overhead budget is useful for working out the pre- determined overhead recovery rates.
- A business may prepare supporting departmental schedules, in which the factory overhead costs are separated into their fixed and variable cost elements. Such schedules enable department managers to direct their attention to those costs for which they are responsible and to evaluate performance of each department.
- A careful study and determination of the behaviour of different types of costs will be essential in preparation of overhead budget.
- A few examples are given below to show how the expenses are estimated.
 1. Fixed expenses are normally policy costs and hence they are based on policy matters.
 2. For estimating indirect labour, work study is resorted to and a

estimate of number of indirect workers required for each level of direct workers employed is made—for example, one supervisor for every twenty direct workers.

3. In regard to the estimate of consumption of indirect materials, the age and condition of the plant and machinery are taken into consideration.

Example of factory overhead budget:

XYZ COMPANY		
Factory overhead budget for the year ending March 31, 20....		
(Anticipated activity of 1,18,000 direct labour hours)		
	(₹)	(₹)
Supplies	12,000	
Indirect labour	30,000	
Cost of fringe benefits	10,000	
Power (variable portion)	22,000	
Maintenance cost (variable portion)	<u>15,000</u>	
Total variable overheads		89,000
Depreciation	10,000	
Property taxes	2,000	
Property insurance	1,000	
Supervision	12,000	
Power (Fixed portion)	800	
Maintenance (Fixed portion)	<u>3,200</u>	
Total fixed overheads		<u>29,000</u>
Total factory overheads		<u>1,18,000</u>
Factory overhead recovery rate is:		
`1,18,000	= `1 per direct labour hour	
1,18,000 labour hours		

(viii) Production Cost Budget:

Production Cost Budget is a forecast of the production for the budget period of an organisation. Production budget is prepared in two parts, viz. production volume budget for the physical units of the products to be manufactured and the cost of production or manufacturing budget detailing the budgeted cost under material, labour, and factory overhead in respect of the products.

Production cost budget covers direct material cost, direct labour cost and manufacturing expenses. After preparing direct material, direct labour and production overhead cost budget, one can prepare production cost budget.

(ix) Ending Inventory Budget:

This budget shows the cost of closing stock of raw materials and finished goods, etc. required to be maintained by the business entity. This information is required to prepare cost-of-goods-sold budget and budgeted financial statements *i.e.*, budgeted income statement and budgeted balance sheet.

Example of end of the year (or closing) inventory budget:

XYZ Company end of the year inventory budget March 31, 20....					
	Units	Unit cost (`)	Amount (`)	Total (`)	
Direct material					
X	3,000	1.50	4,500		
Y	500	2.50	<u>1,250</u>		5,750
Finished goods					
A	500	49.00*	24,500		
B	1,000	53.00*	<u>53,000</u>		<u>77,500</u>
Total					<u>83,250</u>

* Unit cost of finished goods have been computed as below:

	Unit cost of input (`)	Product A		Product B	
		Units	Amount (`)	Units	Amount (`)
Material X	1.50	12	18.00	12.00	18.00
Material Y	2.50	4	10.00	2.00	5.00
Direct labour	2.00	7	14.00	10.00	20.00
Factory overhead	1.00	7	<u>7.00</u>	10.00	<u>10.00</u>
			<u>49.00</u>		<u>53.00</u>

(x) Cost of Goods Sold Budget:

This budget covers direct material cost, direct labour cost and manufacturing expenses. This is adjusted by addition of the cost of the opening inventory and reducing therefrom the cost of closing inventory of finished products.

We present below the cost-of-goods-sold budget on the basis of the data taken from the various budgets already illustrated:

XYZ Company cost-of-goods-sold budget for the year ending March 31, 20....	
	Amount
	(₹)
Direct materials used	3,19,000
Direct labour	2,36,000
Factory overhead	<u>1,18,000</u>
Total manufacturing costs	6,73,000
	<i>Add : Finished goods (opening)</i>
	<u>1,79,50</u>
	<u>0*</u>
	8,52,50
	0
Less : Finished goods (closing)	<u>77,500*</u>
Total cost of goods sold	<u>7,75,000</u>

*Assumed figure

(xi) Selling and Distribution Cost Budget:

Selling and distribution are the essential aspects of the profit earning function. At the same time, the pre-determination of these costs is very difficult. Selling & Distribution Cost Budget is a forecast of the cost of selling & distribution of goods during the budget period.

Selling cost is defined as the cost of seeking to create and stimulate demand and of securing orders. These costs are, therefore, incurred to maintain and increase the level of sales. All expenses connected with advertising, sales promotion, sales office, salesmen, credit collection, market research, after sales service, etc. are generally grouped together to form part of the responsibility of the sales manager.

While making a budget, selling costs are divided into fixed and variable. Semi- variable costs should also be separated into variable and fixed elements.

The problems faced in the preparation of selling cost budgets are:

1. Heavy expenditure on selling and sales promotion may have to be incurred when the volume of sales is falling off. This will increase the percentage of such costs to total sales, and

2. Sometimes intensive sales and promotion efforts are called for in one year and the benefit of such efforts accrue in the subsequent years. This makes it difficult to establish a proportion of selling cost to sales.
3. In spite of these problems, some relationship between selling cost and volume of sales has to be established and it is the duty of the Budget Controller to determine the amount of selling costs to be incurred to achieve the desired level of sales volume.

Using the past experience as a guide, consideration should be given to the future trend of sales, possible changes in competition etc., in pre-determination of selling costs.

- **Distribution cost has been defined as the cost of the sequence of operations which begins with making the packet of product available for dispatch and ends with making the re-conditioned return of empty package, if any available for re-use.** It includes transport cost, storage and warehousing costs, etc.
- Preparation of the advertising cost budget is the responsibility of the sales manager or advertisement manager. When preparing the advertisement cost budget, consideration should be given to the following factors:
 1. The best method of advertisement must be selected; costs will vary according to the method selected.
 2. The maximum amount to be spent in a period, say one year, has to be decided.
 3. Advertising and sales should be co-ordinated. It means that money should be spent on advertisement only when sufficient quantities of the product advertised are ready for sale.
 4. An effective control over advertisement expenditure should be exercised and the effectiveness of the advertisement should be measured.
 5. The choice of the method of advertising a product is based on the effectiveness of the money spent on advertisement in increasing or maintaining sales. If the output sold increases, the production cost will come down because of the economies of large-scale production.

- The amount to be spent on advertisement may be decided on the basis of the following factors:
 1. A percentage on the total sales value of the budget period or on the expected profit may be fixed on the basis of past experience.
 2. A sum which is expected to be incurred by the competitors may be fixed to be spent during the budget period.
 3. A fixed sum per unit of output can be fixed and added to cost.
 4. An amount is fixed on the basis of the ability of the company to spend on advertising.
 5. An advertisement plan is decided upon and the amount to be spent is determined.
- Depending upon the nature of the product and the effectiveness of the media of the advertising the company prepares a schedule of various methods of advertisement, to be used for effective sales promotion. The number of advertisements (insertions) are determined and the cost calculated as per the rates applicable to each of the media selected. This is a sound method.

Example of selling and distribution cost budget:

XYZ Company selling and distribution cost budget for the year ending March 31, 20....	
	Amount
<u>Direct selling expenses:</u>	(`)
Salesmen's salaries	14,500
Salesmen's commission	7,000
Travelling expenses	<u>19,000</u>
	<u>40,500</u>
<u>Distribution expenses:</u>	
Warehouse wages	6,000
Warehouse rent, rates, electricity	4,500
Lorry expenses	<u>11,000</u>
	<u>21,500</u>

Sales office expenses:

Salaries	16,000
Rent, rates, electricity	12,000
Depreciation	2,000
Stationery, postage and telephone	12,500
General expenses	<u>3,000</u>
	<u>45,500</u>

Advertising:

Press	4,500
Radio and television	18,500
Shop window displays	<u>4,000</u>
	<u>27,000</u>
Total	<u>1,34,500</u>

(xii) Administrative Expenses Budget:

The administrative expenses are mostly policy costs and are, therefore, fixed in nature. **The most practical method to follow in preparing estimate of these expenses is to follow the past experience with due regard to anticipated changes either in general policy or the volume of business.** To bring such expenses under control, it is necessary to review them frequently and to determine at regular intervals whether or not these expenses continue to be adjusted. Examples of such expenses are: board meeting expenses, expenditure incurred on staff employed in human resources and finance departments, audit fees, depreciation of office equipment, insurance, subscriptions, postage, stationery, telephone, telegrams, office supplies, etc.

**XYZ Company administrative expenses
budget for the year ending March
31, 20...**

	()
Salaries of clerical staff	28,000
Executives' salaries	8,000
Audit fee	600
Depreciation on office equipment	800
Insurance	250
Stationery	1,250

Postage and telegrams	950
Telephones	850
Miscellaneous	<u>5,300</u>
Total administrative expenses	<u>46,000</u>

(xiii) Research and Development Cost Budget:

Research and development expenditure is to be incurred so that the products or methods of production do not become obsolete. The research and development budget is the forecast of all such expenses. **Research is required in order to develop and/or improve products and methods. When research results in definite benefit to the company, development function begins.** After development, formal production can commence on commercial scale and then production function starts. Since the areas of research and development cannot be precisely defined, the costs incurred under both the functions are clubbed together as research and development costs. Research and Development (R & D) plays a vital role in maintaining the business. For example, automobile manufacturers, and those who produce drugs, spend considerable sums on R & D to improve their products.

Research may be either pure research or applied research. Pure research increases knowledge whereas applied research aims at producing definite results like improved methods of production, etc.

Research and development expenses should be controlled carefully and hence a limit on the spending is placed, *i.e.*, the amount to be spent is carefully determined or allocated.

- The following are the methods of allocation of R & D expenses.
 1. A percentage based on total sales value. This method is good if sales value is steady from year to year.
 2. A percentage based on net profit.
 3. A total sum is estimated on the basis of past experience and future R & D plans and policies.
 4. A sum is fixed on the basis of cash resources available with the company.

All factors which affect the importance of R & D are considered. For example, factors like demand for existing products, competition, economic conditions, etc., are considered carefully and a sum is set aside as R& D budget.

(xiv)Capital Expenditure Budget:

The capital expenditure budget represents the planned outlay on fixed assets like land, building, plant and machinery, etc. during the budget period. This budget is subject to strict management control because it entails large amount of expenditure. The budget is prepared to cover a long period of years and it projects the capital costs over the period in which the expenditure is to be incurred and the expected earnings.

The preparation of capital budget is based on the following considerations:

1. Capital Budget is a budget prepared for capital receipts and expenditure such as investment on land and building, plant and machinery obtaining loans, issue of shares, purchase of assets etc.
2. Future development plans to increase output by expansion of plant facilities.
3. Replacement requests from the concerned departments.
4. Factors like sales potential to absorb the increased output, possibility of price reductions, increased costs of advertising and sales promotion to absorb increased output, etc.
5. Overhead on production facilities of certain departments as indicated by the plant utilisation budget.

Merits/Advantages of capital budgeting

1. Capital budget outlines the capital development programme and estimated capital expenditure during the budget period.
2. It enables the company to establish a system of priorities. When there is a shortage of funds, capital rationing becomes necessary.
3. It serves as a tool for controlling expenditure.

4. It provides the amount of expenditure to be incorporated in the future budget summaries for calculation of estimated return on capital employed.
5. This enables the cash budget to be completed. With other cash commitments capital expenditure commitment should also be considered for the completion of the budget.
6. It facilitates cost reduction programme, particularly when modernisation and renovation is covered by this budget.

ILLUSTRATION 4

A single product company estimated its quarter-wise sales for the next year as under:

Quarter	Sales (Units)
<i>I</i>	30,000
<i>II</i>	37,500
<i>III</i>	41,250
<i>IV</i>	45,000

The opening stock of finished goods is 6,000 units and the company expects to maintain the closing stock of finished goods at 12,250 units at the end of the year. The production pattern in each quarter is based on 80% of the sales of the current quarter and 20% of the sales of the next quarter. The company maintains this 20% of sales of next quarter as closing stock of current quarter.

The opening stock of raw materials in the beginning of the year is 10,000 kg. and the closing stock at the end of the year is required to be maintained at 5,000 kg. Each unit of finished output requires 2 kg. of raw materials.

The company proposes to purchase the entire annual requirement of raw materials in the first three quarters in the proportion and at the prices given below:

Quarter	Purchase of raw materials % to total annual requirement in quantity	Price per kg. (₹)
<i>I</i>	30%	2
<i>II</i>	50%	3
<i>III</i>	20%	4

The value of the opening stock of raw materials in the beginning of the year is ` 20,000. You are required to PREPARE the following for the next year, quarter wise:

- (i) Production budget (in units).
- (ii) Raw material consumption budget (in quantity).
- (iii) Raw material purchase budget (in quantity and value).
- (iv) Priced stores ledger card of the raw material using First in First out method.

SOLUTION

Working Note:

Calculation of total annual production

	(Units)
Sales in 4 quarters	1,53,750
Add: Closing balance	12,250
	1,66,000
Less: Opening balance	(6,000)
Total number of units to be produced in the next year	1,60,000

(i) Production Budget (in units)

Quarters	I Unit s	II Unit s	III Unit s	IV Unit s	Total Unit s
Sales	30,000	37,500	41,250	45,000	1,53,750
Production in current quarter (80% of the sale of current quarter)	24,000	30,000	33,000	36,000	
Production for next quarter (20% of the sale of next quarter)	7,500	8,250	9,000	12,250	
Total production	31,500	38,250	42,000	48,250	1,60,000

(ii) Raw material consumption budget in quantity

Quarters	I	II	III	IV	Total
Units to be produced in each quarter: (A)	31,500	38,250	42,000	48,250	1,60,000
Raw material consumption p.u. (kg.): (B)	2	2	2	2	
Total raw material consumption (Kg.) : (A × B)	63,000	76,500	84,000	96,500	3,20,000

(iii) Raw material purchase budget (in quantity)

	Qty. (kg.)
Raw material required for production	3,20,000
Add : Closing balance of raw material	5,000
	3,25,000
Less : Opening balance	(10,000)
Material to be purchased	3,15,000

Raw material purchase budget (in value)

Quarters	% of annual requirement	Qty. of material	Rate per kg. (₹)	Amount (₹)
(1)	(2)	(3)	(4)	(5)=(3×4)
I	30	94,500 (3,15,000 kg. × 30%)	2	1,89,000
II	50	1,57,500 (3,15,000 kg. × 50%)	3	4,72,500
III	20	63,000 (3,15,000 kg. × 20%)	4	2,52,000
Total		3,15,000		9,13,500

(iv)

**Priced Stores Ledger Card
(of the raw material using FIFO method)**

	Quarters											
	I			II			III			IV		
	Kg.	Rate	Value	Kg.	Rate	Value	Kg.	Rate	Value	Kg.	Rate	Value
		(₹)	(₹)		(₹)	(₹)		(₹)	(₹)		(₹)	(₹)
Opening balance	10,000	2	20,000	41,500	2	83,000	1,22,500	3	3,67,500	38,500	3	1,15,500
(A)										63,000	4	2,52,000
Purchases: (B)	94,500	2	1,89,000	1,57,500	3	4,72,500	63,000	4	2,52,000	-	-	-
Consumption : (C)	63,000	2	1,26,000	41,500	2	83,000	84,000	3	2,52,000	38,500	3	1,15,500
				35,000	3	1,05,000				58,000	4	2,32,000
Balance: (D)	41,500	2	83,000	1,22,500	3	3,67,500	38,500	3	1,15,500	5,000	4	20,000
(D) = (A) + (B) - (C)							63,000	4	2,52,000			

ILLUSTRATION 5

A company is engaged in the manufacture of specialised sub-assemblies required for certain electronic equipment. The company envisages that in the forthcoming month, December, the sales will be in the ratio of 3 : 4 : 2 respectively of sub-assemblies, ACB, MCB and DP.

The following is the schedule of components required for manufacture:

Component requirements					
Sub-assembly	Selling Price	Base board	IC08	IC12	IC26
ACB	520	1	8	4	2
MCB	500	1	2	10	6
DP	350	1	2	4	8
Purchase price (₹)		60	20	12	8

The direct labour time and variable overheads required for each of the sub-assemblies are:

Labour hours Variable overheads (₹)			
	Grade A	Grade B	
ACB	8	16	36
MCB	6	12	24
DP	4	8	24
Direct wage rate per hour (₹)	5	4	—

The labourers work 8 hours a day for 25 days a month.

The opening stocks of sub-assemblies and components for December are as under:

Sub-assemblies		Components	
ACB	800	Base Board	1,600
MCB	1,200	IC08	1,200
DP	2,800	IC12	6,000
		IC26	4,000

Fixed overheads amount to ₹ 7,57,200 for the month and a monthly profit target of ₹ 12 lacs has been set.

The company is eager for a reduction of closing inventories for the month of December of sub-assemblies and components by 10% of quantity as compared to the opening stock. PREPARE the following budgets for the month of December:

- (a) Sales budget in quantity and value.
- (b) Production budget in quantity
- (c) Component usage budget in quantity.
- (d) Component purchase budget in quantity and value.
- (e) Manpower budget showing the number of workers and the amount of wages payable.

SOLUTION

Working Note:

1. Statement showing contribution:

Sub- assemblies	ABC	MCB	DP	Total
	(')	(')	(')	(')
Selling price per unit (p.u.) : (A)	520	500	350	
Marginal Cost per unit.				
Components				
- Base board	60	60	60	
- IC08	160	40	40	
- IC12	48	120	48	
- IC26	16	48	64	
Labour				
- Grade A	40	30	20	
- Grade B	64	48	32	
Variable production overhead	36	24	24	
Total marginal cost per unit. : (B)	424	370	288	

Contribution per unit. : (C) = (A) - (B)	96	130	62	
Sales ratio : (D)	3	4	2	
Contribution × Sales ratio: [(E) = (C) × (D)]	288	520	124	932

2. Desired Contribution for the forthcoming month December

(`)

Fixed overheads 7,57,200

Desired profit 12,00,000

Desired contribution 19,57,200

3. Sales mix required i.e. number of batches for the forthcoming month December

Sales mix required = Desired contribution/contribution × Sales ratio

= `19,57,200/932 (Refer to Working notes 1 and 2)

= 2,100 batches

Budgets for the month of December

(a) Sales budget in quantity and value

Sub-assemblies	ACB	MCB	DP	Total
Sales (Qty.)	6,300 (2,100×3)	8,400 (2,100×4)	4,200 (2,100×2)	
Selling price p.u. (`)	520	500	350	89,46,000
Sales value (`)	32,76,000	42,00,000	14,70,000	

(b) Production budget in quantity

Sub-assemblies	ACB	MCB	DP
Sales	6,300	8,400	4,200
Add : Closing stock	720	1,080	2,520

(Opening stock less 10%)	—	—	—
Total quantity required	7,020	9,480	6,720
Less : Opening stock	(800)	(1,200)	(2,800)
Production	6,220	8,280	3,920

(c) Component usage budget in quantity

Sub-assemblies	ACB	MCB	DP	Total
Production	6,220	8,280	3,920	—
Base board (1 each)	6,220	8,280	3,920	18,420
Component IC08 (8:2:2)	49,760 (6,220 × 8)	16,560 (8,280 × 2)	7,840 (3,920 × 2)	74,160
Component IC12 (4:10:4)	24,880 (6,220 × 4)	82,800 (8,280 × 10)	15,680 (3,920 × 4)	1,23,360
Component IC26 (2:6:8)	12,440 (6,220 × 2)	49,680 (8,280 × 6)	31,360 (3,920 × 8)	93,480

(d) Component Purchase budget in quantity and value

Sub-assemblies	Base board	IC08	IC12	IC26	Total
Usage in production	18,420	74,160	1,23,360	93,480	
Add: Closing stock (Opening stock less 10%)	1,440	1,080	5,400	3,600	
Less: Opening stock	19,860 (1,600)	75,240 (1,200)	1,28,760 (6,000)	97,080 (4,000)	
Purchase (Qty.)	18,260	74,040	1,22,760	93,080	
Purchase price (₹)	60	20	12	8	
Purchase value (₹)	10,95,600	14,80,800	14,73,120	7,44,640	47,94,160

(e) Manpower budget showing the number of workers and the amount of wages payable

Sub-assemblies	Budgeted Production	Direct labour				Total
		Grade A		Grade B		
		Hours p.u.	Total hours	Hours p.u.	Total hours	
ACB	6,220	8	49,760	16	99,520	
MCB	8,280	6	49,680	12	99,360	
DP	3,920	4	15,680	8	31,360	
(A) Total hours			1,15,120		2,30,240	
(B) Hours per man per month			200		200	
(C) Number of workers per month: (A/B)			576		1,152	
(D) Wage rate per month (`)			1,000		800	
(E) Wages payable (`) : (C × D)			5,76,000		9,21,600	14,97,600

(xv) Cash Budget:

Cash Budget is a detailed budget of cash receipts and cash payments incorporating both revenue and capital items for the budget period. This budget is usually of two parts giving detailed estimates of (i) cash receipts and (ii) cash disbursements. Estimates of cash-receipts are prepared on a monthly basis and depend upon estimated cash-sales, collections from debtors and anticipated receipts from other sources such as sale of assets, borrowings, etc. Estimates of cash disbursements are based on estimated cash purchases, payments to creditors, employees' remuneration, bonus, advances to suppliers, budgeted capital expenditure for expansion, etc.

Cash budget represents the cash requirements of the business during the budget period. **It is the plan of receipts and payments of cash for the budget period**, analysed to show the monthly flow of cash drawn up in such a way that the balance can be forecasted at regular intervals.

The cash budget is one of the most important elements of the budgeted balance sheet. Information from the various operating budgets, such as the sales budget, the direct materials purchases budget, and the selling and administrative expenses budget, affects the cash budget.

In addition, the capital expenditures budget, dividend policies, and plans for equity or long-term debt financing also affect the cash budget.

The main objectives of preparing cash budget are:

- (i) The probable cash position, as a result of planned operation, is assessed; and thus, the excess or shortage of cash becomes clear. This helps in arranging short-term borrowings in advance to meet the situations of shortage of cash or making investments when cash is in excess.
- (ii) Cash can be coordinated in relation to total working capital, sales investment and debt.
- (iii) A sound basis for credit for current control of cash position is established.
- (iv) The effect of sudden and seasonal requirements, large stocks, delay in collection of receipts, etc., on the cash position of the organization is revealed and things become under to the management.

Advantages of cash budget

- (i) It aids in securing option working capital need for smooth running of the operation and planning for payments to the shareholders.
- (ii) It eases strains of a cash shortage
- (iii) It facilitates temporary cash investment wherever, and to whatever extent, found in excess
- (iv) It provides for normal growth

8.3 Master Budget

CIMA, London, defines it as “the summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed.” When all the necessary functional budgets have been prepared, the budget officer will prepare the master budget which may consist of budgeted profit and loss account and budgeted balance sheet. These are in fact the budget summaries. When the master budget is approved by the board of directors, it represents a standard for the achievement of which all the departments will work. On the basis of the various

budgets (schedules) prepared earlier in this study, we prepare income statement and budgeted balance sheet. below budgeted

Example of budgeted income statement:

XYZ Company Budgeted Income Statement For the Year Ending March 31, 20....		
	(₹)	Amount (₹)
Sales		11,75,000
Less: Cost of goods sold		<u>7,75,000</u>
Gross margin		4,00,000
Less: Selling and distribution expenses	1,36,500	
Less: Administrative expenses	<u>46,000</u>	<u>1,82,500</u>
Profit before interest and taxes		2,17,500
Interest expenses (assumed)		<u>50,000</u>
Profit before tax		1,67,500
Income-tax (30% assumed)		<u>50,250</u>
Net profit		<u>1,17,250</u>

Example of budgeted balance sheet:

XYZ Company Budgeted Balance Sheet March 31, 20....		
	(₹)	(₹)
Share capital	3,50,000	
Retained income	<u>1,29,000</u>	4,79,000
<i>Represented by:</i>		
Plant and machinery	3,40,000	
Less: Provision for depreciation	<u>60,000</u>	2,80,000
Raw materials	5,750	
Finished goods	77,500	
Debtors	1,10,000	
Cash	<u>37,750</u>	2,31,000
Less: Creditors		<u>32,000</u>
		<u>4,79,000</u>
Note: Information not available in respect of share capital, opening balance of retained earnings, current assets and current liabilities, etc., has been assumed to complete the above balance sheet.		

ILLUSTRATION 6

Float glass Manufacturing Company requires you to PREPARE the Master budget for the next year from the following information:

Sales:

Toughened Glass ` 6,00,000

Bent Glass ` 2,00,000

Direct material cost *60% of sales*

Direct wages 20 workers @ ` 150 per month

Factory overheads:

Indirect labour -

Works manager ` 500 per month

Foreman ` 400 per month

Stores and spares 2.5% on sales

Depreciation on machinery ` 12,600

Light and power ` 3,000

Repairs and maintenance ` 8,000

Others sundries 10% on direct wages

Administration, selling and distribution expenses ` 36,000 per year

SOLUTION

Master Budget for the year ending _____

Sales:			(`)
Toughened Glass			6,00,000
Bent Glass			2,00,000
Total Sales			8,00,000
Less: Cost of production:			
Direct materials (60% of `8,00,000)		4,80,000	

Direct wages (20 workers × ₹150 × 12months)		36,000	
Prime Cost		5,16,000	
Fixed Factory Overhead:			
Works manager's salary (500 × 12)	6,000		
Foreman's salary (400 × 12)	4,800		
Depreciation	12,600		
Light and power (assumed fixed)	3,000	26,400	
Variable Factory Overhead:			
Stores and spares	20,000		
Repairs and maintenance	8,000		
Sundry expenses	3,600	31,600	
Works Cost			5,74,000
Gross Profit (Sales - Works cost)			2,26,000
Less: Adm., selling and distribution expenses			36,000
Net Profit			1,90,000

8.4 Classification on the basis of Time Period

These types of Budgets are classified on the basis of time periods. These types of budgets reflect the planning period of the organization.

Long term Budget: - Long Term Budget is a budget prepared covering a period of more than a year. The Budgets are prepared to depict long term planning of the business. The period of long-term Budgets varies between three to ten years. These budgets are useful for those industries where gestation period is long i.e., the business entities manufacturing machinery, electricity etc.

1. **Short term Budget:** - These budgets are generally for one or two years and are in the form of monetary terms. The consumer's good industries like Sugar, Cotton, and textile use short term budgets.

2. **Current Budgets:** - The period of current budgets is generally of months and weeks. These budgets relate to the current activities of the business. According to CIMA London "Current budget is a budget which is created which is established for use over a short period of time and is related to current conditions".

9. ZERO – BASED BUDGETING (ZBB)

Zero-based Budgeting (ZBB) is defined as a method of budgeting which requires each cost element to be specifically justified, though the activities to which the budget relates are not being undertaken for the first time. The cost of each activity has to be justified and without justification, the budget allowance is zero.

Zero based budgeting differs from the conventional system of budgeting because it mainly starts from scratch or zero and not on the basis of trends or historical levels of expenditure. In the customary budgeting system, the last year's figures are accepted as they are, or cut back or increases are granted. Zero based budgeting on the other hand, starts with the premise that the budget for next period is zero so long the demand for a function, process, project or activity is not justified for each rupee from the first rupee spent.

Zero-based Budgeting (ZBB) is an emergent form of budgeting which arises to overcome the limitations of incremental (traditional) budgeting system.

ZBB is an activity-based budgeting system where budgets are prepared for each activity rather than functional department. Justification in the form of cost benefits for the activity is required to be given. The activities are then evaluated and prioritized by the management on the basis of factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

ZBB is suitable for both corporate and non-corporate entities. In case of non- corporate entities like Government department, local bodies, not for profit organisations, where these entities need to justify the benefits of expenditures on social programmes like mid-day meal, installation of street lights, provision of drinking water etc.

In case of corporate entities, ZBB is best suited for discretionary costs like research and development cost, training programmes, advertisement etc.

9.1 Stages in Zero-based Budgeting

ZBB involves the following stages:

- (i) Identification and description of Decision packages
- (ii) Evaluation of Decision packages
- (iii) Ranking (Prioritisation) of the Decision packages
- (iv) Allocation of resources

(i) Identification and description of Decision packages: Decision packages are the programmes or activities for which decision is required to be taken. The programmes or activities are described for technical specifications, financial impact in the form of cost benefit analysis and other issues like environmental, regulatory, social etc.

(ii) Evaluation of Decision packages: Once Decision packages are identified and described, it is evaluated against factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

(iii) Ranking (Prioritisation) of the Decision packages: After evaluation of the decision packages, it is ranked on the basis priority of the activities. Because of this prioritization feature **ZBB is also known as Priority-based Budgeting.**

(iv) Allocation of resources: After ranking of the decision packages, resources are allocated for decision packages. Budgets are prepared like it is done first time without taking reference to previous budgets.

9.2 Advantages of Zero-based Budgeting

The advantages of zero-based budgeting are as follows:

- It **provides a systematic approach for the evaluation of different activities** and rank them in order of preference for the allocation of scarce resources.
- It **ensures that the various functions undertaken by the organization are critical** for the achievement of its objectives and are being performed in the best possible way.

- It **provides an opportunity to the management to allocate resources** for various activities only after having a thorough cost-benefit-analysis. The chances of arbitrary cuts and enhancement are thus avoided.
- The **areas of wasteful expenditure can be easily identified and eliminated.**
- Departmental budgets are closely linked with corporation objectives.
- **The technique can also be used for the introduction and implementation of the system of 'management by objective.'** Thus, it cannot only be used for fulfillment of the objectives of traditional budgeting but it can also be used for a variety of other purposes.

Zero-based budgeting is superior to traditional budgeting: Zero based budgeting is superior to traditional budgeting in the following manner:

- It provides a systematic approach for evaluation of different activities.
- It ensures that the function undertaken are critical for the achievement of the objectives.
- It provides an opportunity for management to allocate resources to various activities after a thorough – cost benefit analysis.
- It helps in the identification of wasteful expenditure and then their elimination. It facilitates the close linkage of departmental budgets with corporate objectives
- It helps in the introduction of a system of Management by Objectives.

9.3 Difference between Traditional Budgeting and Zero- based budgeting

Following are the points of difference between traditional budgeting and zero- based budgeting:

- Traditional budgeting is accounting oriented. Main stress happens to be on previous level of expenditure. Zero-based budgeting makes a decision- oriented approach. It is very rational in nature and requires all programmes, old and new, to compete for scarce resources.

- In traditional budgeting, first reference is made to past level of spending and then demand for inflation and new programmes. In zero-based budgeting, management focuses attention to only on decision packages, which enjoy priority to others.
- In tradition budgeting, some managers deliberately inflate their budget request so that after the cuts they still get what they want. In zero-based budgeting, a rationale analysis of budget proposals is attempted. The managers, who unnecessarily try to inflate the budget request, are likely to be caught and exposed. Management accords its approval only to a carefully devised result-oriented package.
- Traditional budgeting is not as clear and as responsive as zero-base budgeting.
- In traditional budgeting, it is for top management to decide why a particular amount should be spent on a particular decision unit. In Zero-based budgeting, this responsibility is shifted from top management to the manager of decision unit.
- Traditional budgeting makes a routine approach. Zero-based budgeting makes a very straightforward approach and immediately spotlights the decision packages enjoying priority over others.

9.4 Limitations of Zero-based Budgeting

- The work involves in the creation of decision-making and their subsequent ranking has to be made on the basis of new data. This process is very tedious to management.
- The activities selected for the purpose of ZBB are on the basis of the traditional functional departments. So, the consideration scheme may not be implemented properly.



10. PERFORMANCE BUDGETING (PB)

Performance budgeting (PB) involves evaluation of the performance of an organisation in the context of both specific as well as overall objectives of the organisation. This requires complete clarity about both the short-term as well as

long-term organisational objectives. The responsibility of the various levels of management should be predetermined in terms of results expected from them and the authority vested in them. In other words, performance budgeting requires fixing of the responsibility of each executive in organisation and the continuous appraisal of his performance. It is, therefore, considered to be synonymous with responsibility accounting.

Performance Budgeting provide a meaningful relationship between estimated inputs and expected outputs as an integral part of the budgeting system. **A performance budget is one which presents the purposes and objectives for which funds are required,** the costs of the programmes proposed for achieving those objectives, and quantitative data measuring the accomplishments and work performed under each programme. Thus, PB is a technique of presenting budgets for costs and revenues in terms of functions. Programmes and activities are correlating the physical and financial aspect of the individual items comprising the budget.

10.1 Traditional Budgeting vs. Performance Budgeting

- The traditional budgeting gives more emphasis on the financial aspect than the physical aspects or performance. PB aims at establishing a relationship between the inputs and the outputs.
- Traditional budgets are generally prepared with the main basis towards the objects or items of expenditure i.e., it highlights the items of expenditure, namely, salaries, stores and materials, rates, rents and taxes and so on. In the PB emphasis is more on the functions of the organisation, the programmes to discharge this function and the activities which will be involved in undertaking these programmes.

10.2 Steps in Performance Budgeting

According to the Administrative Reforms Commission (ARC, the following steps are the basic ones in PB:

- Establishing a meaningful functional programme and activity classification of government operations.

- Bring the system of accounting and financial management in accordance with this classification.
- Evolving suitable norms, yardsticks, work units of performance and units costs, wherever possible under each programme and activity for their reporting and evaluation.

The Report of the ARC use the following terms in an integrated sequence:



The team 'function' is used in the sense of 'objective'. For achieving objectives 'programmes' will have to be evolved. In respect of time horizon, it is essentially a replacement of traditional annual fiscal budgeting by a more output-oriented, but still an annual, exercise.

For an enterprise that wants to adopt PB, it is thus imperative that:

- the objectives of the enterprise are spelt out in concrete terms.
- the objectives are then translated into specific functions, programmes, activities and tasks for different levels of management within the realities of fiscal; constraints;
- realistic and acceptable norms, yardsticks or standards and performance indicators should be evolved and expressed in quantifiable physical units.
- a style of management based upon decentralised responsibility structure should be adopted, and
- an accounting and reporting system should be developed to facilities monitoring, analysis and review of actual performance in relation to budgets.

Performance Reporting at various levels of management:

Report : A major part of the management accountant's job consists of preparing reports to provide information for purposes of control and planning.

The important consideration in drawing up of reports and determining their scope are the following:

Significance either	:	Are the facts in the reports reliable? Does it called for action or demonstrate the effect of action? It is material enough.
Timeliness use?	:	How late can the information be and still be of use? What is the earliest moment at which it could be used if it were available? How frequently is it required?
Accuracy not	:	How small should be an inaccuracy which does alter the significance of the information?
Appropriateness that	:	Is the recipient the right person to take any action is needed? Is there any other information which is required to support the information to anyone else jointly interested?
Discrimination of	:	Will anything be lost by omitting the item? Will any the items gain from the omission? Is the responsibility for suppressing the item acceptable?
Presentation it is	:	Is the report clear and unbiased? Is the form of it suitable to the subject? Is the form of it suitable to the recipient?

The following are certain types of reports which are to be prepared and submitted to management regularly at predetermined time interval:

- 1. Top Management:** (Including Board of Directors and financial managers)
 - (i) Balance Sheet
 - (ii) Profit & Loss Statement
 - (iii) Position of stocks
 - (iv) Disposition of funds or working capital;
 - (v) Capital expenditure and forward commitments together with progress of projects in hands;
 - (vi) Cash-flow statements;

(vii) Sales, production, and other appropriate statistics.

2. Sales Management:

- (i) Actual sales compared with budgeted sales to measure performance by:
 - Products,
 - Territories
 - Individual salesmen, and
 - Customers.
- (ii) Standard profit and loss by product:
 - For fixing selling prices, and
 - To Concentrate on sales of most profitable products.
- (iii) Selling expenses in relation to budget and sales value analyzed by:
 - Products,
 - Territories
 - Individual salesmen, and
 - Customers.
- (iv) Bad debts and accounts which are slow and difficult in collection.
- (v) Status reports on new or doubtful customers.

3. Production Management:

- (i) To Buyer: Price variations on purchases analysed by commodities.
- (ii) To Foreman:
 - Operational efficiency for individual operators duly summarized as departmental average;
 - Labour utilization report and causes of lost time and controllable time;
 - Indirect shop expenses against the standard allowed; and
 - Scrap report.
- (iii) To Works Managers:
 - Departmental operating statement;

- General works operating statements (Expenses relating to all works expenses not directly allocable or controllable by departments);
- Plant utilization report;
- Department Scrap report; and
- Material usage report.

4. Special Reports:

These reports may be prepared at the request of general management or at the initiative of the management accountants. The necessity for them may, in some cases, arise on account of the need for more detailed information on matters of interest first revealed; by the routine, reports. These reports may range over a very wide area. Some of the matters in respect of which such reports may be required can be:

- (i) Taxation legislation and its effect on profits.
- (ii) Estimates of the earning capacity of a new project.
- (iii) Break-even analysis
- (iv) Replacement of capital equipment.
- (v) Special pricing analysis
- (vi) Make or buy certain components
- (vii) Statement of surplus available for payment of bonus under the labour appellate tribunal formula.

15.11. BUDGET RATIO

Ratio is a mathematical relationship between two or more related figures. Budget ratios provide information about the performance level, i.e., the extent of deviation of actual performance from the budgeted performance and whether the actual performance is favourable or unfavorable. If the ratio is 100% or more, the performance is considered as favourable and if ratio is less than 100% the performance is considered as unfavourable.

The following ratios are usually used by the management to measure development from budget.

Capacity Usage Ratio: This relationship between the budgeted number of working hours and the maximum possible number of working hours in a budget period.

Standard Capacity Employed Ratio: This ratio indicates the extent to which facilities were actually utilized during the budget period.

Level of Activity Ratio: This may be defined as the number of standard hours equivalent to work produced expressed as a percentage of the budget of standard hours.

Efficiency Ratio: This ratio may be defined as standard hours equivalent of work produced expressed as a percentage of the actual hours spent in producing the work.

Calendar Ratio: This ratio may be defined as the relationship between the number of working days in a period and the number of working as in the relative budget period.

Budget

$$(i) \quad \text{Efficiency Ratio} = \frac{\text{Standard}}{\text{Actual}}$$

$$(ii) \quad \text{Activity} = \frac{\text{Standard Hours}}{\text{Budgeted}}$$

$$(iii) \quad \text{Calendar Ratio} = \frac{\text{Available working days}}{\text{Budgeted working}}$$

$$(iv) \quad \text{Standard Capacity Usage} = \frac{\text{Budgeted Hours}}{\text{Max. possible hours in the budgeted}}$$

$$(v) \quad \text{Actual Capacity Usage Ratio} = \frac{\text{Max. possible working hours in a}}{\text{Max. possible working hours in a}} \times 10$$

$$(vi) \quad \text{Actual Usage of Budgeted Capacity Ratio} = \frac{\text{Actual working}}{\text{Budgeted}}$$

ILLUSTRATION 7

Following data is available for DKG and Co:

Standard working hours 8 hours per day of 5 days per
week Maximum capacity 50 employees
Actual working 40 employees
Actual hours expected to be worked per four week
6,400 hours Std. hours expected to be earned per four
weeks 8,000 hours Actual hours worked in the four-
week period 6,000
hours Standard hours earned in the four- week period
7,000 hours.

The related period is of 4 weeks. In this period there was a one special day holiday due to national event. CALCULATE the following ratios:

(1) Efficiency Ratio, (2) Activity Ratio, (3) Calendar Ratio, (4) Standard Capacity Usage Ratio, (5) Actual Capacity Usage Ratio. (6) Actual Usage of Budgeted Capacity Ratio.

SOLUTION

Maximum Capacity in a budget period

= 50 Employees × 8 Hrs. × 5 Days × 4 Weeks = 8,000

Hrs. Budgeted Hours

40 Employees × 8 Hrs. × 5 Days × 4 Weeks = 6,400

Hrs. Actual Hrs. = 6,000 Hrs. (given)

Standard Hrs. for Actual Output = 7,000 Hrs.

Budget No. of Days = 20 Days = 20 Days (4 Weeks × 5 Days)

Actual No. of Days = 20 - 1 = 19 Days

1. **Efficiency Ratio** = $\frac{\text{Standard Hrs}}{\text{Actual Hrs}} \times 100 = \frac{7,000}{6,000} \times 100 = 116.67\%$
2. **Activity Ratio** = $\frac{\text{Standard Hrs}}{\text{Budgeted Hrs}} \times 100 = \frac{7,000}{6,400} \times 100 = 109.375\%$

$$\begin{aligned}
 \text{3. Calendar Ratio} &= \frac{\text{Available working days}}{\text{Budgeted working days}} \times 100 = \frac{19 \text{ days}}{20 \text{ days}} \times 100 = 95\%
 \end{aligned}$$

$$\begin{aligned}
 \text{4. Standard Capacity Usage Ratio} &= \frac{\text{Budgeted Hours}}{\text{Max. possible hours in the budgeted period}} \times 100 = \frac{6,400 \text{ hours}}{8,000 \text{ hours}} \times 100 = 80\%
 \end{aligned}$$

$$\begin{aligned}
 \text{5. Actual Capacity Usage Ratio} &= \frac{\text{Actual Hoursworked}}{\text{Max. possible working hours in a period}} \times 100 \\
 &= \frac{6,000 \text{ hours}}{8,000 \text{ hours}} \times 100 = 75\%
 \end{aligned}$$

$$\begin{aligned}
 \text{6. Actual Usage of Budgeted Capacity Ratio} &= \frac{\text{Actual working Hours}}{\text{Budgeted Hours}} \times 100 \\
 &= \frac{6,000 \text{ hours}}{6,400 \text{ hours}} \times 100 = 93.75\%
 \end{aligned}$$

SUMMARY

- ♦ **Budget:** Budget is a quantitative expression of a plan of action to be pursued over a defined period of time. It is statement of an estimated performance to be achieved in given time, expressed in monetary or quantitative or both terms.
- ♦ **Budget Centre:** A Budget Centre is a section of an organisation developed for the purpose of budgetary control, and is intended to facilitate formulation of various budgets with the help of head of the department.
- ♦ **Budgetary Control:** Budgetary Control is the establishment of budgets, relating the responsibilities of executives to the requirements of a policy, and the continuous comparison of actual with budgeted results, either to secure by individual action the objective of that policy or to provide a basis for its revision.
- ♦ **Budget Manual:** The Budget manual is a document or booklet which Contain guidelines for the preparation of budget in an organization.
- ♦ **Budget Period:** The period of time for which a budget is prepared and used. It may be a year, quarter or a month.

Capacity based	-	Fixed and Flexible
Content based	-	Monetary and Physical/quantitative
Functional based Administrative,	-	Purchase, Sale, Production Cost, Selling & Distribution, Research & Development, Plant Capital Expenditure, Cash, Plant Utilization.

- ♦ **Fixed Budget:** a fixed budget, is a budget designed to remain unchanged irrespective of the level of activity actually attained
- ♦ **Flexible Budget:** a flexible budget is defined as a budget which, by recognizing the difference between fixed, semi-variable and variable costs is designed to change in relation to the level of activity attained.
- ♦ **Zero-based Budgeting (ZBB):** Zero-based Budgeting (ZBB) is defined as 'a method of budgeting which requires each cost element to be specifically justified, although the activities to which the budget relates are not being undertaken for the first time, without approval, the budget allowance is zero
- ♦ **Performance Budgeting (PB):** Performance Budgeting means that budget in which the responsibility of various levels of management is predetermined in terms of output or result keeping in view the authority vested with them.
- ♦ **Budget Ratios:** Ratio is a mathematical relationship between two or more related figures. Budget ratios provide information about the performance level, i.e., the extent of deviation of actual performance from the budgeted performance and whether the actual performance is favourable or unfavorable.

TEST YOUR KNOWLEDGE

Multiple Choice Questions (MCQs)

1. *If a company wishes to establish a factory overhead budget system in which estimated costs can be derived directly from estimates of activity levels, it should prepare a:*
 - (a) Master budget
 - (b) Cash budget

- (c) *Flexible budget*
 - (d) *Fixed budget*
2. *The classification of fixed and variable cost is useful for the preparation of:*
- (a) *Master budget*
 - (b) *Flexible budget*
 - (c) *Cash budget*
 - (d) *Capital budget*
3. *Budget manual is a document:*
- (a) *Which contains different type of budgets to be formulated only.*
 - (b) *Which contains the details about standard cost of the products to be made.*
 - (c) *Setting out the budget organization and procedures for preparing a budget including fixation of responsibilities, formats and records required for the purpose of preparing a budget and for exercising budgetary control system.*
 - (d) *None of the above*
4. *The budget control organization is usually headed by a top executive who is known as:*
- (a) *General manager*
 - (b) *Budget director/budget controller*
 - (c) *Accountant of the organization*
 - (d) *None of the above*
5. *"A favourable budget variance is always an indication of efficient performance".*
Do you agree, give reason?
- (a) *A favourable variance indicates, saving on the part of the organization hence it indicates efficient performance of the organization.*
 - (b) *Under all situations, a favourable variance of an organization speaks about its efficient performance.*

- (c) *A favourable variance does not necessarily indicate efficient performance, because such a variance might have been arrived at by not carrying out the expenses mentioned in the budget.*
 - (d) *None of the above.*
6. *A budget report is prepared on the principle of exception and thus-*
- (a) *Only unfavourable variances should be shown*
 - (b) *Only favourable variance should be shown*
 - (c) *Both favourable and unfavourable variances should be shown*
 - (d) *None of the above*
7. *Purchases budget and materials budget are same:*
- (a) *Purchases budget is a budget which includes only the details of all materials purchased*
 - (b) *Purchases budget is a wider concept and thus includes not only purchases of materials but also other item's as well*
 - (c) *Purchases budget is different from materials budget; it includes purchases of other items only*
 - (d) *None of the above*
8. *Efficiency ratio is:*
- (a) *The extent of actual working days avoided during the budget period*
 - (b) *Activity ratio/ capacity ratio*
 - (c) *Whether the actual activity is more or less than budgeted activity*
 - (d) *None of the above*
9. *Activity Ratio depicts:*
- (a) *Whether actual capacity utilized exceeds or falls short of the budgeted capacity*
 - (b) *Whether the actual hours used for actual production were more or less than the standard hours*
 - (c) *Whether actual activity was more or less than the budgeted capacity*

- (d) None of the above
10. Which of the following is usually a short-term budget:
- (a) Capital expenditure budget
 - (b) Research and development budget
 - (c) Cash budget
 - (d) Sales budget

Theoretical Questions

1. EXPLAIN briefly the concept of 'flexible budget'.
2. DISCUSS the components of budgetary control system.
3. LIST the eight functional budgets prepared by a business.
4. DISTINGUISH between Fixed and flexible budget.
5. EXPLAIN the Essentials of budget.
6. STATE the considerations on which capital expenditure budget is prepared.
7. DESCRIBE the steps involved in the budgetary control technique.
8. DESCRIBE the salient features of budget manual.

Practical Problems

1. B Ltd manufactures two products viz., X and Y and sells them through two divisions, East and West. For the purpose of Sales Budget to the Budget Committee, following information has been made available for the year 2022-23:

Product	Budgeted Sales		Actual Sales	
	East Division	West Division	East Division	West Division
X	800 units at '18	1,200 units at '18	1,000 units at '18	1,400 units at '18
Y	600 units at '42	1,000 units at '42	400 units at '42	800 units at '42

Adequate market studies reveal that product X is popular but underpriced. It is expected that if the price of X is increased by ` 2, it will, find a ready market. On the other hand, Y is overpriced and if the price of Y is reduced by ` 2 it will have more demand in the market. The company management has agreed for the aforesaid price changes. On the basis of these price changes and the reports of salesmen, following estimates have been prepared by the Divisional Managers:

Percentage increase in sales over budgeted sales

Product	East Division	West Division
X	+ 12.5%	+ 7.5%
Y	+ 22.5%	+ 12.5%

With the help of intensive advertisement campaign, following additional sales (over and above the above-mentioned estimated sales by Divisional Managers) are possible:

Product	East Division	West Division
X	120 units	140 units
Y	80 units	100 units

You are required to PREPARE Sales Budget for 2022-23 after incorporating above estimates and also SHOW the Budgeted Sales and Actual Sales of 2021-22.

- During the FY 2021-22, P Limited has produced 60,000 units operating at 50% capacity level. The cost structure at the 50% level of activity is as under:

	(`)
Direct Material	300 per unit
Direct Wages	100 per unit
Variable Overheads	100 per unit
Direct Expenses	60 per unit
Factory Expenses (25% fixed)	80 per unit
Selling and Distribution Exp. (80% variable)	40 per unit
Office and Administrative Exp. (100% fixed)	20 per unit

The company anticipates that in FY 2022-23, the variable costs will go up by 20% and fixed costs will go up by 15%.

The selling price per unit will increase by 10% to ` 880

Required:

- (i) CALCULATE the budgeted profit/ loss for the FY 2021-22.
 - (ii) PREPARE an Expense budget on marginal cost basis for the FY 2022-23 for the company at 50% and 60% level of activity and FIND OUT the profits at respective levels.
3. K Ltd. produces and markets a very popular product called 'X'. The company is interested in presenting its budget for the second quarter of 2022-23. The following information are made available for this purpose:

- (i) It expects to sell 1,50,000 bags of 'X' during the second quarter of 2022- 23 at the selling price of ` 1,200 per bag.
- (ii) Each bag of 'X' requires 2.5 mtr. of raw - material 'Y' and 7.5 mtr. of raw - material 'Z'.
- (iii) Stock levels are planned as follows:

Particulars	Beginning of Quarter	End of Quarter
Finished Bags of 'X' (Nos.)	45,000	33,000
Raw - Material 'Y' (mtr)	96,000	78,000
Raw - Material 'Z' (mtr)	1,71,000	1,41,000
Empty Bag (Nos.)	1,11,000	84,000

- (iv) 'Y' cost `160 per mtr., 'Z' costs `30 per mtr. and 'Empty Bag' costs `110 each.
- (v) It requires 9 minutes of direct labour to produce and fill one bag of 'X'.
Labour cost is ` 70 per hour.
- (vi) Variable manufacturing costs are ` 60 per bag. Fixed manufacturing costs ` 40,00,000 per quarter.

(vii) Variable selling and administration expenses are 5% of sales and fixed administration and selling expenses are ` 3,75,000 per quarter.

Required

- (i) PREPARE a production budget for the said quarter in quantity.
 - (ii) PREPARE a raw - material purchase budget for 'Y', 'Z' and 'Empty Bags' for the said quarter in quantity as well as in rupees.
 - (iii) COMPUTE the budgeted variable cost to produce one bag of 'X'.
4. ABC Ltd. is currently operating at 75% of its capacity. In the past two years, the levels of operations were 55% and 65% respectively. Presently, the production is 75,000 units. The company is planning for 85% capacity level during 2022-
23. The cost details are as follows:

	55%	65%	75%
	(`)	(`)	(`)
Direct Materials	11,00,000	13,00,000	15,00,000
Direct Labour	5,50,000	6,50,000	7,50,000
Factory Overheads	3,10,000	3,30,000	3,50,000
Selling Overheads	3,20,000	3,60,000	4,00,000
Administrative Overheads	<u>1,60,000</u>	<u>1,60,000</u>	<u>1,60,000</u>
	<u>24,40,000</u>	<u>28,00,000</u>	<u>31,60,000</u>

Profit is estimated @ 20% on sales.

The following increases in costs are expected during the year:

In percentage

Direct Materials	8
Direct Labour	5
Variable Factory Overheads	5
Variable Selling Overheads	8
Fixed Factory Overheads	10
Fixed Selling Overheads	15
Administrative Overheads	10

PREPARE flexible budget for the period 2022-23 at 85% level of capacity. Also ascertain profit and contribution.

5. The accountant of manufacturing company provides you the following details for year 2021-22:

	(₹)		(₹)
Direct materials	1,75,000	Other variable costs	80,000
Direct Wages	1,00,000	Other fixed costs	80,000
Fixed factory overheads	1,00,000	Profit	1,15,000
Variable factory overheads	1,00,000	Sales	7,50,000

During the year, the company manufactured two products A and B and the output and costs were:

	A	B
Output (units)	2,00,000	1,00,000
Selling price per unit	₹ 2.00	₹ 3.50
Direct materials per unit	₹ 0.50	₹ 0.75
Direct wages per unit	₹ 0.25	₹ 0.50

Variable factory overhead is absorbed as a percentage of direct wages. Other variable costs have been computed as: Product A ₹ 0.25 per unit; and B ₹ 0.30 per unit.

During 2022-23, it is expected that the demand for product A will fall by 25 % and for B by 50%. It is decided to manufacture a further product C, the cost for which is estimated as follows:

	Product C
Output (units)	2,00,000
Selling price per unit	₹ 1.75
Direct materials per unit	₹ 0.40
Direct wages per unit	₹ 0.25

It is anticipated that the other variable costs per unit will be the same as for product A.

PREPARE a budget to present to the management, showing the current position and the position for 2022-23. Comment on the comparative results.

6. *TQM Ltd. has furnished the following information for the month ending 30th June:*

	Master Budget	Actual	Variance
<i>Units produced and sold</i>	80,000	72,000	
<i>Sales (₹)</i>	3,20,000	2,80,000	40,000 (A)
<i>Direct material (₹)</i>	80,000	73,600	6,400 (F)
<i>Direct wages (₹)</i>	1,20,000	1,04,800	15,200 (F)
<i>Variable overheads (₹)</i>	40,000	37,600	2,400 (F)
<i>Fixed overhead (₹)</i>	40,000	39,200	800 (F)
<i>Total Cost</i>	2,80,000	2,55,200	

The Standard costs of the products are as follows:

	Per unit (₹)
<i>Direct materials (1 kg. at the rate of ₹1 per kg.)</i>	1.00
<i>Direct wages (1 hour at the rate of ₹1.50)</i>	1.50
<i>Variable overheads (1 hour at the rate of ₹0.50)</i>	0.50

Actual results for the month showed that 78,400 kg. of material were used and 70,400 labour hours were recorded.

Required:

- (i) PREPARE Flexible budget for the month and compare with actual results.*
 - (ii) CALCULATE Material, Labour, Variable Overhead and Fixed Overhead Expenditure variances.*
7. *Jigyasa Ltd. is drawing a production plan for its two products Minimax (MM) and Heavyhigh (HH) for the year 2022-23. The company's policy is to hold closing stock of finished goods at 25% of the anticipated volume of sales of the*

succeeding month. The following are the estimated data for two products:

	Minimax (MM)	Heavyhigh (HH)
Budgeted Production units	1,80,000	1,20,000
	(₹)	(₹)
Direct material cost per unit	220	280
Direct labour cost per unit	130	120
Manufacturing overhead	4,00,000	5,00,000

The estimated units to be sold in the first four months of the year 2022-23 are as under

	April	May	June	July
Minimax	8,000	10,000	12,000	16,000
Heavyhigh	6,000	8,000	9,000	14,000

PREPARE production budget for the first quarter in month-wise.

8. Concorde Ltd. manufactures two products using two types of materials and one grade of labour. Shown below is an extract from the company's working papers for the next month's budget:

	Product - A	Product - B
Budgeted sales (in units)	2,400	3,600
Budgeted material consumption per unit (in kg):		
Material-X	5	3
Material-Y	4	6
Standard labour hours allowed per unit of product	3	5

Material-X and Material-Y cost ₹ 4 and ₹ 6 per kg and labours are paid

₹ 25 per hour. Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 180 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%. In

addition the non-productive down-time is budgeted at 20% of the productive hours worked.

There are four 5-days weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period .

It is anticipated that stock at the beginning of the period will be:

Product-A	400 units
Product-B	200 units
Material-X	1,000 kg.
Material-Y	500 kg.

The anticipated closing stocks for budget period are as below:

Product-A	4 days sales
Product-B	5 days sales
Material-X	10 days consumption
Material-Y	6 days consumption

Required:

CALCULATE the Material Purchase Budget and the Wages Budget for the direct workers, showing the quantities and values, for the next month.

Case Scenarios

1. M Ltd. is a public sector undertaking (PSU), produces a product A. The company is in process of preparing its revenue budget for the year 2024. The company has the following information which can be useful in preparing the budget:
 - (i) It has anticipated 12% growth in sales volume from the year 2023 of 4,20,000 tonnes.
 - (ii) The sales price of ` 23,000 per tonne will be increased by 10% provided Wholesale Price Index (WPI) increases by 5%.
 - (iii) To produce one tonne of product A, 2.3 tonnes of raw material are required. The raw material cost is ` 4,500 per tonne. The price of raw material will also increase by 10% if WPI increase by 5%.
 - (iv) The projected increase in WPI for 2022 is 4%

- (v) A total of 6,000 employees works for the company. The company works 26 days in a month.
- (vi) 85% of employees of the company are permanent and getting salary as per 5- year wage agreement. The earnings per manshift (means an employee cost for a shift of 8 hours) is ` 3,000 (excluding terminal benefits). The new wage agreement will be implemented from 1st July 2024 and it is expected that a 15% increase in pay will be given.
- (vii) The casual employees are getting a daily wage of ` 850. The wages are linked to Consumer Price Index (CPI). The present CPI is 165.17 points and it is expected to be 173.59 points in year 2024.
- (viii) Power cost for the year 2021 is ` 42,00,000 for 7,00,000 units (1 unit = 1 Kwh). 60% of power is used for production purpose (directly related to production volume) and remaining are for employee quarters and administrative offices.
- (ix) During the year 2023, the company has paid ` 60,00,000 for safety and maintenance works. The amount will increase in proportion to the volume of production.
- (x) During the year 2023, the company has paid ` 1,20,000 for the purchase of diesel to be used in car hired for administrative purposes. The cost of diesel will increase by 15% in year 2024.
- (xi) During the year 2023, the company has paid ` 6,00,000 for car hire charges (excluding fuel cost). In year 2024, the company has decided to reimburse the diesel cost to the car rental company. Doing this will attract 5% GST on Reverse Charge Mechanism (RCM) basis on which the company will not get GST input credit.
- (xii) Depreciation on fixed assets for the year 2023 is ` 80,40,00,000 and it will be 15% lower in 2024.

You being an associate to the budget controller of the company is expected to answer the following question:

- (i) What would be the sales volume for the FY 2024?
 - (a) 4,70,400 tonnes
 - (b) 4,70,000 tonnes
 - (c) 4,70,600 tonnes
 - (d) 4,70,200 tonnes

- (ii) What would be quantity of raw material in FY 2024?
t
- (a) 9,66,000 tonnes
 - (b) 1,81,000 tonnes
 - (c) 10,81,900 tonnes
 - (d) 10,81,920 tonnes
- (iii) What would be the car hire charges for the FY 2023?
t
- (a) ₹ 6,00,000
 - (b) ₹ 6,50,000
 - (c) ₹ 6,40,000
 - (d) ₹ 6,20,000
- (iv) What would be the car hire charges for the FY 2024?
t
- (a) ₹ 6,00,000
 - (b) ₹ 7,74,900
 - (c) ₹ 6,83,000
 - (d) ₹ 6,20,000
- (v) What would be the budgeted profit/ loss for the year 2024?
t
- (a) ₹ 1273.043 lakhs
 - (b) (₹ 5142 lakhs)
 - (c) ₹ 5142 lakhs
 - (d) (₹ 1273.043 lakhs)

ANSWERS/ SOLUTIONS

Answers to the MCQs

1.	(c)	2.	(b)	3.	(c)	4.	(b)	5.	(c)	6.	(c)
7.	(b)	8.	(b)	9.	(c)	10.	(c)				

Answers to the Theoretical Questions

1. Please refer paragraph 8.1
2. Please refer paragraph 5.7
3. Please refer paragraph 8.2
4. Please refer paragraph 8.1
5. Please refer paragraph 2
6. Please refer paragraph 8.2
7. Please refer paragraph 5.3
8. Please refer paragraph 7

Answers to the Practical Problems

1. Statement Showing Sales Budget for 2022-23

Division	Product X			Product Y			Total
	Qty.	Rate (`)	Amt. (`)	Qty.	Rate (`)	Amt. (`)	Amt. (`)
East	1,020 ₁	20	20,400	815 ³	40	32,600	53,000
West	1,430 ₂	20	28,600	1,225 ⁴	40	49,000	77,600
Total	1,200		49,000	1,000		81,600	1,30,600

Workings

1. $800 \times 112.5\% + 120 = 1,020$ units
2. $1,200 \times 107.5\% + 140 = 1,430$ units
3. $600 \times 122.5\% + 80 = 815$ units
4. $1,000 \times 112.5\% + 100 = 1,225$ units

Statement Showing Sales Budget for 2021-22

Division	Product X			Product Y			Total
	Qty.	Rate (`)	Amt. (`)	Qty.	Rate (`)	Amt. (`)	Amt. (`)
East	800	18	14,400	600	42	25,200	39,600
West	1,200	18	21,600	1,000	42	42,000	63,600
Total	2,000		36,000	1,600		67,200	1,03,200

Statement Showing Actual Sales for 2021-22

Division	Product X			Product Y			Total
	Qty.	Rate (`)	Amt. (`)	Qty.	Rate (`)	Amt. (`)	Amt. (`)
East	1,000	18	18,000	400	42	16,800	34,800
West	1,400	18	25,200	800	42	33,600	58,800
Total	2,400		43,200	1,200		50,400	93,600

2. (i) Calculation of Budgeted profit for the FY 2021-22

	60,000 units	
	Per unit (`)	Amount (`)
Sales (A)	800.00	4,80,00,000
Variable Costs:		
- Direct Material	300.00	1,80,00,000
- Direct Wages	100.00	60,00,000
- Variable Overheads	100.00	60,00,000
- Direct Expenses	60.00	36,00,000
- Variable factory expenses (75% of `80 p.u.)	60.00	36,00,000
- Variable Selling & Dist. exp. (80% of `40 p.u.)	32.00	19,20,000
Total Variable Cost (B)	652.00	3,91,20,000
Contribution (C) = (A - B)	148.00	88,80,000

Fixed Costs:			
- Office and Admin. exp. (100%)		--	12,00,000
- Fixed factory exp. (25%)		--	12,00,000
- Fixed Selling & Dist. exp. (20%)		--	4,80,000
Total Fixed Costs	(D)	--	28,80,000
Profit	(C - D)	--	60,00,000

(ii) Expense Budget of P Ltd. for the FY 2022-23 at 50% & 60% level

	60,000 units		72,000 units	
	Per unit (`)	Amount (`)	Per unit (`)	Amount (`)
Sales (A)	880.00	5,28,00,000	880.00	6,33,60,000
Variable Costs:				
- Direct Material	360.00	2,16,00,000	360.00	2,59,20,000
- Direct Wages	120.00	72,00,000	120.00	86,40,000
- Variable Overheads	120.00	72,00,000	120.00	86,40,000
- Direct Expenses	72.00	43,20,000	72.00	51,84,000
- Variable	72.00	43,20,000	72.00	51,84,000
factory expenses				
- Variable Selling & Dist. exp.	38.40	23,04,000	38.40	27,64,800
Total Variable Cost (B)	782.40	4,69,44,000	782.40	5,63,32,800
Contribution (C) = (A - B)	97.60	58,56,000	97.60	70,27,200
Fixed Costs:				
- Office and Admin. exp. (100%)	--	13,80,000	--	13,80,000
- Fixed factory exp. (25%)	--	13,80,000	--	13,80,000
- Fixed Selling & Dist. exp. (20%)	--	5,52,000	--	5,52,000

Total Fixed Costs (D)	--	33,12,000	--	33,12,000
Profit (C - D)	--	25,44,000	--	37,15,200

3. (i) Production Budget of 'X' for the Second Quarter

Particulars	Bags (Nos.)
Budgeted Sales	1,50,000
Add: Desired Closing stock	33,000
Total Requirements	1,83,000
Less: Opening stock	(45,000)
Required Production	1,38,000

(ii) Raw-Materials Purchase Budget in Quantity as well as in ` for 1,38,000 Bags of 'X'

Particulars	'Y' Mtr.	'Z' Mtr.	Empty Bags Nos.
Production Requirements Per bag of 'X'	2.5	7.5	1.0
Requirement for Production	3,45,000	10,35,000	1,38,000
	(1,38,000 × 2.5)	(1,38,000 × 7.5)	(1,38,000 × 1)
Add:	78,000	1,41,000	84,000
Desired Closing Stock			
Total Requirements	4,23,000	11,76,000	2,22,000
Less:	(96,000)	(1,71,000)	(1,11,000)
Opening Stock			
Quantity to be purchased	3,27,000	10,05,000	1,11,000
Cost per mtr./Bag	`160	`30	`110
Cost of Purchase (`)	5,23,20,000	3,01,50,000	1,22,10,000

(iii) Computation of Budgeted Variable Cost of Production of 1 Bag of 'X'

Particulars	(`)
Raw - Material	
Y 2.5 mtr @160	400.00
Z 7.5 mtr @30	225.00
Empty Bag	110.00
Direct Labour (`70× 9 minutes / 60 minutes)	10.50
Variable Manufacturing Overheads	60.00
Variable Cost of Production per bag	805.50

4.

ABC Ltd.

Budget for 85% capacity level for the period 2022-23

Budgeted production (units)		85,000
	Per Unit (`)	Amount (`)
Direct Material (note 1)	21.60	18,36,000
Direct Labour (note 2)	10.50	8,92,500
Variable factory overhead (note 3)	2.10	1,78,500
Variable selling overhead (note 4)	4.32	3,67,200
Variable cost	38.52	32,74,200
Fixed factory overhead (note 3)		2,20,000
Fixed selling overhead (note 4)		1,15,000
Administrative overhead		1,76,000
Fixed cost		5,11,000
Total cost		37,85,200
Add: Profit 20% on sales or 25% on total cost		9,46,300
Sales		47,31,500
Contribution (Sales - Variable cost)		14,57,300

Working Notes:

1. Direct Materials:

	(₹)		(₹)
75% Capacity	15,00,000	65% Capacity	13,00,000
65% Capacity	13,00,000	55% Capacity	11,00,000
10% change in capacity	2,00,000	10% change in capacity	2,00,000

For 10% increase in capacity, i.e., for increase by 10,000 units, the total direct material cost regularly changes by ₹ 2,00,000

Direct material cost (variable) = ₹ 2,00,000 ÷ 10,000 = ₹ 20

After 8% increase in price, direct material cost per unit = ₹ 20 × 1.08
= ₹ 21.60

Direct material cost for 85,000 budgeted units = 85,000 × ₹ 21.60
= ₹ 18,36,000

2. Direct Labour:

	(₹)		(₹)
75% Capacity	7,50,000	65% Capacity	6,50,000
65% Capacity	6,50,000	55% Capacity	5,50,000
10% change in capacity	1,00,000	10% change in capacity	1,00,000

For 10% increase in capacity, direct labour cost regularly changes by ₹ 1,00,000.

Direct labour cost per unit = ₹ 1,00,000 ÷ 10,000 = ₹ 10

After 5% increase in price, direct labour cost per unit = ₹ 10 × 1.05
= ₹ 10.50

Direct labour for 85,000 units = 85,000 units × ₹ 10.50 = ₹ 8,92,500.

3. Factory overheads are semi-variable overheads:

	(₹)		(₹)
75% Capacity	3,50,000	65% Capacity	3,30,000
65% Capacity	3,30,000	55% Capacity	3,10,000
10% change in capacity	20,000	10% change in capacity	20,000

Variable factory overhead = ₹ 20,000 ÷ 10,000 = ₹ 2

Variable factory overhead for 75,000 units = 75,000 × ₹ 2 =

₹ 1,50,000 Fixed factory overhead = ₹ 3,50,000 - ₹ 1,50,000 = ₹ 2,00,000.

Variable factory overhead after 5% increase = ₹ 2 × 1.05 = ₹ 2.10

Fixed factory overhead after 10% increase = ₹ 2,00,000 × 1.10 = ₹ 2,20,000.

4. Selling overhead is semi-variable overhead:

	(₹)		(₹)
75% Capacity	4,00,000	65% Capacity	3,60,000
65% Capacity	3,60,000	55% Capacity	3,20,000
10% change in capacity	40,000	10% change in capacity	40,000

Variable selling overhead = ₹ 40,000 ÷ 10,000 units = ₹ 4

Variable selling overhead for 75,000 units = 75,000 × ₹ 4 = ₹

3,00,000. Fixed selling overhead = ₹ 4,00,000 - ₹ 3,00,000 = ₹ 1,00,000

Variable selling overhead after 8% increase = ₹ 4 × 1.08 = ₹ 4.32

Fixed selling overhead after 15% increase = ₹ 1,00,000 × 1.15 = ₹ 1,15,000

5. Administrative overhead is fixed:

After 10% increase = ₹ 1,60,000 × 1.10 = ₹ 1,76,000

5.

Budget Showing Current Position and Position for 2022-23

	Position for 2021-22			Position for 2022-23			
	A	B	Total (A+B)	A	B	C	Total (A+B+C)
Sales (units)	2,00,000	1,00,000	-	1,50,000	50,000	2,00,000	-
	(`)	(`)	(`)	(`)	(`)	(`)	(`)
(A) Sales	4,00,000	3,50,000	7,50,000	3,00,000	1,75,000	3,50,000	8,25,000
Direct Material	1,00,000	75,000	1,75,000	75,000	37,500	80,000	1,92,500
Direct wages	50,000	50,000	1,00,000	37,500	25,000	50,000	1,12,500
Factory overhead (variable)	50,000	50,000	1,00,000	37,500	25,000	50,000	1,12,500
Other	50,000	30,000	80,000	37,500	15,000	50,000	1,02,500
variable costs							
(B) Marginal Cost	2,50,000	2,05,000	4,55,000	1,87,500	1,02,500	2,30,000	5,20,000
(C) Contribution (A-B)	1,50,000	1,45,000	2,95,000	1,12,500	72,500	1,20,000	3,05,000
Fixed costs – Factory			1,00,000				1,00,000
– Others			80,000				80,000
(D) Total fixed cost			1,80,000				1,80,000
Profit (C – D)			1,15,000				1,25,000

Comments: Introduction of Product C is likely to increase profit by ` 10,000 (i.e. from ` 1,15,000 to ` 1,25,000) in 2022-23 as compared to 2021-22. Therefore, introduction of product C is recommended.

6. (i) Statement showing Flexible Budget and its comparison with actual

		Master Budget 80,000 units	Flexible Budget (at standard cost)		Actual for 72,000 units	Variance
			Per unit	72,000 units		
A.	Sales	3,20,000	4.00	2,88,000	2,80,000	8,000 (A)
B.	Direct material	80,000	1.00	72,000	73,600	1,600 (A)
C.	Direct wages	1,20,000	1.50	1,08,000	1,04,800	3,200 (F)
D.	Variable overhead	40,000	0.50	36,000	37,600	1,600 (A)
E.	Total variable cost	2,40,000	3.00	2,16,000	2,16,000	~
F.	Contribution	80,000	1.00	72,000	64,000	~
G.	Fixed overhead	40,000	0.50	40,000	39,200	800 (F)
H.	Net profit	40,000	0.50	32,000	24,800	7,200 (A)

(ii) Variances:

$$\begin{aligned}
 \text{Direct Material Cost Variance} &= \text{Standard Cost for Actual output} \\
 &\quad - \text{Actual cost} \\
 &= ₹ 72,000 - ₹ 73,600 = ₹ 1,600 \text{ (A)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Direct Material Price Variance} &= \text{Actual Quantity (Standard rate} \\
 &\quad - \text{Actual Rate)} \\
 &= 78,400 \text{ units} \times ₹ 1.00 - ₹ 73,600 \\
 &\quad \text{②} \quad \text{③} \quad \text{④} \\
 &\quad \text{⑤} \\
 &= ₹ 4,800 \text{ (F)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Direct Material Usage Variance} &= \text{Standard Rate (Std. Qty. - Actual Quantity)} \\
 &= ₹ 1 (72,000 \text{ units} - 78,400 \text{ units}) \\
 &= ₹ 6,400 \text{ (A)}
 \end{aligned}$$

Direct Labour Cost Variance = Standard Cost for actual output - Actual cost

$$= ₹ 1,08,000 - ₹ 1,04,800 = ₹ 3,200 \text{ (F)}$$

Direct Labour Rate Variance = Actual Hour (Std Rate - Actual Rate)

$$= \frac{70,400 \text{ hours}}{1.5} \times ₹ 1.5 - ₹ 1,04,800$$

$$= ₹ 800 \text{ (F)}$$

Direct Labour Efficiency = Standard Rate (Standard Hour - Actual Hour)

$$= ₹ 1.5 (72,000 - 70,400) = ₹ 2,400 \text{ (F)}$$

Variable Overhead = Recovered variable overhead - Actual variable overhead

$$= (72,000 \text{ units} \times ₹ 0.50) - ₹ 37,600$$

$$= ₹ 1,600 \text{ (A)}$$

Fixed Overhead Expenditure = Budgeted fixed overhead - Actual fixed overhead

$$= ₹ 40,000 - ₹ 39,200 = ₹ 800 \text{ (F)}$$

7. Production Budget of Product Minimax and Heavyhigh (in units)

	April		May		June		Total	
	MM	HH	MM	HH	MM	HH	MM	HH
Sales	8,000	6,000	10,000	8,000	12,000	9,000	30,000	23,000
Add: Closing Stock (25% of next month's sale)	2,500	2,000	3,000	2,250	4,000	3,500	9,500	7,750
Less: Opening Stock	2,000*	1,500*	2,500	2,000	3,000	2,250	7,500	5,750
Production units	8,500	6,500	10,500	8,250	13,000	10,250	32,000	25,000

* Opening stock of April is the closing stock of March, which is as per company's

policy

25% of next month'' sale.

Production Cost Budget

Element of cost	Rate (₹)		Amount (₹)	
	MM (32,000 units)	HH (25,000 units)	MM	HH
Direct Material	220	280	70,40,000	70,00,000
Direct Labour	130	120	41,60,000	30,00,000
Manufacturing Overhead (4,00,000 ÷ 1,80,000 × 32,000) (5,00,000 ÷ 1,20,000 × 25,000)			71,111	1,04,167
			1,12,71,111	1,01,04,167

8. Number of days in budget period = 4 weeks × 5 days = 20 days

Number of units to be produced

	Product-A (units)	Product-B (units)
Budgeted Sales	2,400	3,600
Add: Closing stock $\frac{2,400 \text{ units}}{20 \text{ days}} \times 4 \text{ days}$ $\frac{3,600 \text{ units}}{20 \text{ days}} \times 5 \text{ days}$	480	900
Less: Opening stock	400	200
	2,480	4,300

(i) Material Purchase Budget

	Material-X (Kg.)	Material-Y (Kg.)
Material required: Product-A	12,400 (2,480 units × 5 kg.)	9,920 (2,480 units × 4 kg.)

Product-B	12,900 (4,300 units × 3 kg.)	25,800 (4,300 units × 6 kg.)
	25,300	35,720
Add: Closing stock  $\frac{25,300\text{kgs.}}{20\text{days}} \times 10\text{days}$  $\frac{35,720\text{kgs.}}{20\text{days}} \times 6\text{days}$	12,650	10,716
Less: Opening stock	1,000	500
Quantity to be purchased	36,950	45,936
Rate per kg. of Material	` 4	` 6
Total Cost	` 1,47,800	` 2,75,616

(ii) Wages Budget

	Product-A (Hours)	Product-B (Hours)
Units to be produced	2,480 units	4,300 units
Standard hours allowed per unit	3	5
Total Standard Hours allowed	7,440	21,500
Productive hours required for production	$\frac{7,440\text{hours}}{80\%} = 9,300$	$\frac{21,500\text{hours}}{80\%} = 26,875$
Add: Non-Productive down time	1,860 hours. (20% of 9,300 hours)	5,375 hours. (20% of 26,875 hours)
Hours to be paid	11,160	32,250

Total Hours to be paid = 43,410 hours (11,160 + 32,250)

Hours to be paid at normal rate = 4 weeks × 40 hours × 180 workers
= 28,800 hours

Hours to be paid at premium rate = 43,410 hours - 28,800 hours = 14,610 hours

Total wages to be paid = 28,800 hours × ₹ 25 + 14,610 hours × ₹ 37.5

= ₹ 7,20,000 + ₹ 5,47,875

= ₹ 12,67,875

Answers To The Case Scenarios

1.

i.	(a)	ii.	(d)	iii.	(a)	iv.	(b)	v.	(d)
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(i) (a) 4,70,400 tonnes

(ii) (d) 10,81,920 tonnes

(iii) (a) ₹ 6,00,000

(iv) (b) ₹ 7,74,900

(v) (d) (₹1273.043 lakhs)

Revenue Budget (Flexible Budget) of M Ltd. for the Year 2024

	Particulars	PY 2023	CY 2024
A	Sales Volume (Tonnes)	4,20,000	4,70,400 [112%×4,20,000]
B	Selling Price per tonne (₹)	23,000	23,000
		(₹ in lakh)	(₹ in lakh)
C	Sales value [A×B]	96,600	1,08,192
D	Raw material Cost:		
(i)	Qty. of Material [2.3 tonnes × A] (tonnes)	9,66,000	10,81,920
(ii)	Price per tonne (₹)	4,500	4,500

(iii)	Total raw material cost (` in lakh) [(i)×(ii)]	43,470	48,686.40
E	Wages & Salary Cost:		
(i)	Wages to casual employees (15%×6,000 = 900 employees)	2,386.80 [900×26×12×`850]	2,508.47 [900×26×12×`893.33]
(ii)	Salary to permanent employees (85%×6,000 = 5,100 employees)	47,736 [5100×26×12×`3,000]	51,316.20 [(5100×26×6×`3,000)+ (5100×26×6×`3,450)]
(iii)	Total wages & salary [(i)+(ii)+(iii)]	50,122.80	53,824.67
F	Power cost:		
(i)	For production (units)	4,20,000 [60%×7,00,000]	4,70,400 [112%×4,20,000]
(ii)	For employees & offices (units) [40%×7,00,000]	2,80,000	2,80,000
(iii)	Total Power consumption (units) [(i)+(ii)]	7,00,000	7,50,400
(iv)	Power rate per unit (`) [`42,00,000÷7,00,000]	6.00	6.00
(v)	Total power cost [(iii)×(iv)]	42	45.024
G	Safety and maintenance Cost	60	67.20 [112%×4,20,000]

H	Diesel cost	1.2	-
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I	Car Hire charge:		
(i)	Car hire charge	6	6
(ii)	Fuel reimb ursement cost	-	1.38 [115%×1.2]
(iii)	GST@5% on RCM basis [5%×(i+ii)]	-	0.369
(iv)	Total Car hire charge cost [(i)+(ii)+(iii)]	6	7.749
J	Depreciation	8,040	6,834 [85%×8040]
K	Total Cost [Sum of D to J]	1,01,742	1,09,465.043
L	Profit/ (Loss) [C-L]	(5,142)	(1,273.043)